Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**CALCULATION** **3420 PRESSURE OXDN.- AUTOCL. BLDG.** **DESIGN OF SCRUBBER SUPPORT FDN.**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
AUTOCL.	:	Autoclave
OXDN.	:	Oxidation
FDN.	:	Foundation
BLDG.	:	Building
PRESS.	:	Pressure

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1. INTENT

This document is to prepare design of foundation for **Scrubber Support Structure**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

Design drawings are attached in Attachment - 1.For load data document no. MD-216-3420-EG-ST-CAL-2242 is to be referred. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 475 Kn/m^2 at 3.0 m below EGL. (Refer Table 5 - BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-Z	Along-X
2	0.70m	0.70m

Size of fdn. raft:	4.00m	4.00m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.70	m
U/S level of fdn:	-3.00	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

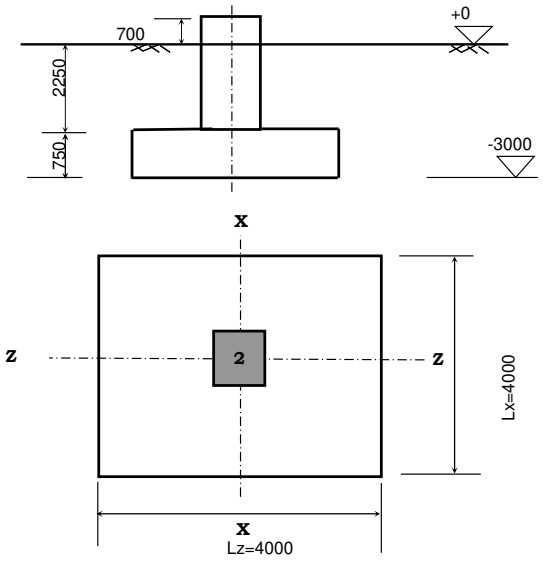
Weight of raft = (4*4*0.75*25) = 300.00KN

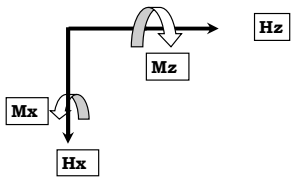
Weight of soil = ((4*4)-(0.7*0.7))*2.25*20 = 697.95KN

Total = 997.95KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	190.54	25.52	-94.41	-0.21	-0.77
202 1.0DL + 1.0LL	569.68	84.03	-310.92	1.72	6.35
203 1.0DL + 1.0WLX1	108.49	-64.03	236.92	3.80	14.07
204 1.0DL + 1.0WLX2	272.59	115.07	-425.74	-4.22	-15.61
205 1.0DL + 1.0WLZ1	552.32	42.02	-155.47	-103.37	-382.48
206 1.0DL + 1.0WLZ2	-171.24	9.01	-33.34	102.96	380.94
207 1.0DL + 0.75WLX1 + 0.75LL	413.35	2.24	-8.29	4.24	15.70
208 1.0DL + 0.75WLX2 + 0.75LL	536.43	136.57	-505.29	-1.77	-6.56
209 1.0DL + 0.75WLZ1 + 0.75LL	746.23	81.78	-302.59	-76.14	-281.72
210 1.0DL + 0.75WLZ2 + 0.75LL	203.56	57.03	-210.99	78.61	290.85
211 0.6DL + 1.0WLX1	32.27	-74.24	274.68	3.89	14.38
212 0.6DL + 1.0WLX2	196.38	104.86	-387.97	-4.14	-15.30
213 0.6DL + 1.0WLZ1	476.10	31.81	-117.71	-103.29	-382.17
214 0.6DL + 1.0WLZ2	-247.45	-1.19	4.42	103.04	381.25

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	1188.5	-0.21	-0.8	25.5	-94.4	25.2	3087.0	18.6	2285.6
202 1.0DL + 1.0LL	1567.6	1.72	6.3	84.0	-310.9	10.1	494.1	7.5	365.6
203 1.0DL + 1.0WLX1	1106.4	3.80	14.1	-64.0	236.9	9.3	157.3	6.9	116.4
204 1.0DL + 1.0WLX2	1270.5	-4.22	-15.6	115.1	-425.7	6.0	162.8	4.4	120.5
205 1.0DL + 1.0WLZ1	1550.3	-103.37	-382.5	42.0	-155.5	19.9	8.1	14.8	6.0
206 1.0DL + 1.0WLZ2	826.7	102.96	380.9	9.0	-33.3	49.6	4.3	36.7	3.2
207 1.0DL + 0.75WLX1 + 0.75LL	1411.3	4.24	15.7	2.2	-8.3	340.4	179.8	251.9	133.0
208 1.0DL + 0.75WLX2 + 0.75LL	1534.4	-1.77	-6.6	136.6	-505.3	6.1	467.5	4.5	346.0
209 1.0DL + 0.75WLZ1 + 0.75LL	1744.2	-76.14	-281.7	81.8	-302.6	11.5	12.4	8.5	9.2
210 1.0DL + 0.75WLZ2 + 0.75LL	1201.5	78.61	290.9	57.0	-211.0	11.4	8.3	8.4	6.1
211 0.6DL + 1.0WLX1	1030.2	3.89	14.4	-74.2	274.7	7.5	143.3	5.6	106.0
212 0.6DL + 1.0WLX2	1194.3	-4.14	-15.3	104.9	-388.0	6.2	156.1	4.6	115.5
213 0.6DL + 1.0WLZ1	1474.1	-103.29	-382.2	31.8	-117.7	25.0	7.7	18.5	5.7
214 0.6DL + 1.0WLZ2	750.5	103.04	381.2	-1.2	4.4	339.7	3.9	251.4	2.9

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	100.03	0
Min Px	2	201	0	100.03	0
Max Py	587	209	0	178.27	0
Min Py	574	214	0	1.02	0
Max Pz	2	201	0	100.03	0
Min Pz	2	201	0	100.03	0

Gross Maximum Base pressure, p_{max} = 178.3 KN/m^2

Net Allowable Safe Bearing Capacity = 475 KN/m^2

Depth of soil above footing = 2.25 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 520 KN/m^2

> 178.3 KN/m^2

Hence OK

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Design of foundation raft F1

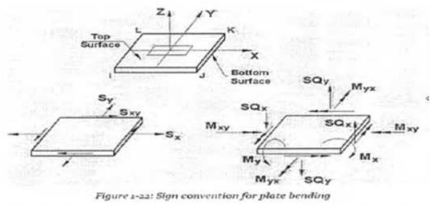
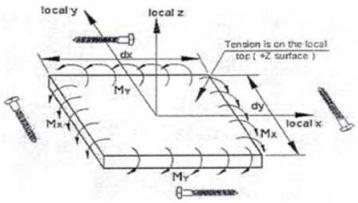


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	347	0.554	-0.309	0	0	0	-109.9	-195.37	-21.16	-131.1	-216.53	105:1.4DL + 1.6WLZ1 + 1.0 LL
Min Qx	327	-1.087	-0.344	0	0	0	-412.1	-147.02	-61.927	-474.07	-208.95	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Qy	414	-0.499	0.708	0	0	0	-237.6	-165.08	72.525	-310.14	-237.6	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Qy	345	-0.219	-1.177	0	0	0	-198.9	-381.59	-62.124	-260.99	-443.71	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Sx	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sx	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Sy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Sxy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sxy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Mx	427	-0.356	-0.196	0	0	0	242.35	-104.72	86.443	328.791	-191.16	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mx	327	-1.087	-0.344	0	0	0	-412.1	-147.02	-61.927	-474.07	-208.95	104:1.4DL + 1.6WLX2 + 1.0 LL
Max My	345	0.228	0.693	0	0	0	80.747	272.058	45.401	126.148	317.459	110:0.9DL + 1.6WLZ2
Min My	345	-0.219	-1.177	0	0	0	-198.9	-381.59	-62.124	-260.99	-443.71	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Mxy	417	-0.112	-0.075	0	0	0	7.193	20.45	157.969	165.162	178.419	105:1.4DL + 1.6WLZ1 + 1.0 LL
Min Mxy	347	-0.04	0.247	0	0	0	72.151	-0.945	-122.34	194.487	-123.28	104:1.4DL + 1.6WLX2 + 1.0 LL

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	378	0.139	-0.141	0	0	0	-70.9	-50.41	-31.189	-102.09	-81.599	105:1.4DL + 1.6WLZ1 + 1.0 LL
Min Qx	341	-0.268	-0.038	0	0	0	-61.27	-106.61	3.434	-64.705	-110.05	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Qy	375	0.024	0.169	0	0	0	63.478	40.276	-1.704	65.182	41.98	110:0.9DL + 1.6WLZ2
Min Qy	375	-0.002	-0.285	0	0	0	-133.5	-54.427	-9.125	-142.58	-63.552	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Sx	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sx	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Sy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Sxy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Min Sxy	588	-0.003	-0.002	0	0	0	-0.551	-0.47	0.206	-0.757	-0.676	101:1.4DL
Max Mx	375	0.024	0.169	0	0	0	63.478	40.276	-1.704	65.182	41.98	110:0.9DL + 1.6WLZ2
Min Mx	375	-0.002	-0.285	0	0	0	-133.5	-54.427	-9.125	-142.58	-63.552	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max My	351	0.093	-0.013	0	0	0	9.325	46.107	11.936	21.261	58.043	110:0.9DL + 1.6WLZ2
Min My	341	-0.259	0.016	0	0	0	-45.85	-127.12	-14.63	-60.477	-141.75	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Mxy	387	0.08	0.023	0	0	0	-30.57	-12.785	35.733	-66.302	-48.518	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mxy	377	0.083	-0.029	0	0	0	-32.06	-14.541	-35.741	-67.796	-50.282	104:1.4DL + 1.6WLX2 + 1.0 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

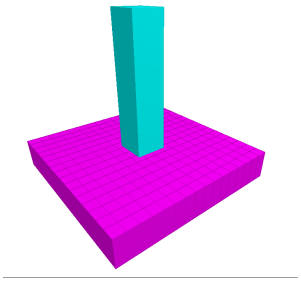
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.4D	328.79	top face	50	692	328.79	0.92	0.0022	1532.3	1245.6	16	131.27	125	1609.2	22.7	28.36	0.07	tension control
104:1.4D	-474.07	bottom face	50	690	474.07	1.33	0.0032	2232.1		20	140.8	125	2514.3	35.5	44.31	0.044	tension control
105:1.4D	165.16	top face	50	692	165.16	0.46	0.0011	763.6		16	161.48	125	1609.2	22.7	28.36	0.07	tension control
104:1.4D	194.49	top face	50	692	194.49	0.54	0.0013	900.5		16	161.48	125	1609.2	22.7	28.36	0.07	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.139	1	104.25	102.09	0.002	0.692	674.36	682.385	1187.24	674.36	2532	> 104.25	ok	505.8	φVc>Vu	not required
	Min Qx	0.268	1	201	64.705			713.835			682.38	2538	> 201	ok	511.8	φVc>Vu	not required

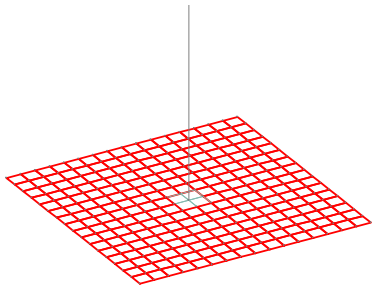
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
110:0.9D	317.46	top face	50	692	317.46	0.88	0.0021	1478.7	1245.6	16	136.03	125	1609.2	22.7	28.36	0.07	tension control
105:1.4D	-443.71	bottom face	50	690	443.71	1.24	0.0030	2086.0		20	150.67	125	2514.3	35.5	44.31	0.044	tension control
105:1.4D	178.42	top face	50	692	178.42	0.50	0.0012	825.5		16	161.48	125	1609.2	22.7	28.36	0.07	tension control
104:1.4D	-123.28	bottom face	50	692	123.28	0.34	0.0008	568.9		16	161.48	125	1609.2	22.7	28.36	0.07	tension control

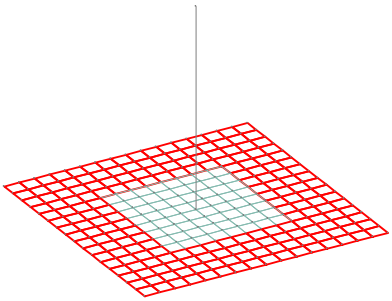
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.169	1	126.75	41.98	0.002	0.692	712.186	682.385	fc'1187	682.38	714	> 126.75	ok	511.8	φVc>Vu	not required
	Min Qy	0.285	1	213.75	63.552			718.699			682.38	714	> 213.75	ok	511.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	20φT	125	c/c
At Top	16φT	125	c/c
Along Z Direction :			
At Bottom	20φT	125	c/c
At Top	16φT	125	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-Z	Along-X
2	0.70m	0.70m

Size of fdn. raft:	4.00m	4.50m
Thickness of raft:	0.75	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.70	m
U/S level of fdn:	-1.25	m
Unit weight of conc:	25	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

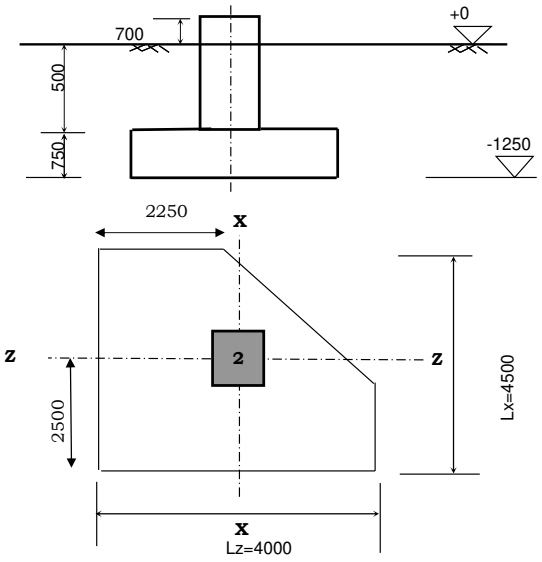
Weight of raft = (4*2.5+ 0.5*(4+2)*2)*25 = 406.25KN

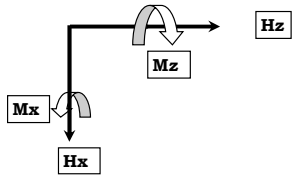
Weight of soil = [{(4*2.5+ 0.5*(4+2)*2)]-(0.7*0.7)]*0.5*20 = 157.60KN

Total = 563.85KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of fooring	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	96.02	-9.66	18.84	-1.33	-2.59
202 1.0DL + 1.0LL	359.60	-40.11	78.20	-3.51	-6.85
203 1.0DL + 1.0WLX1	224.04	-76.53	149.22	-5.34	-10.41
204 1.0DL + 1.0WLX2	-31.99	57.20	-111.53	2.68	5.22
205 1.0DL + 1.0WLZ1	323.60	-16.38	31.94	-98.00	-191.10
206 1.0DL + 1.0WLZ2	-131.56	-2.95	5.75	-98.00	-191.10
207 1.0DL + 0.75WLX1 + 0.75LL	389.71	-82.64	161.15	-5.97	-11.64
208 1.0DL + 0.75WLX2 + 0.75LL	197.69	17.65	-34.42	0.04	0.08
209 1.0DL + 0.75WLZ1 + 0.75LL	464.39	-37.53	73.19	-75.47	-147.17
210 1.0DL + 0.75WLZ2 + 0.75LL	123.02	-27.46	53.54	-75.47	-147.17
211 0.6DL + 1.0WLX1	185.63	-72.66	141.68	-4.80	-9.37
212 0.6DL + 1.0WLX2	-70.40	61.06	-119.07	3.21	6.26
213 0.6DL + 1.0WLZ1	285.20	-12.52	24.41	-97.47	-190.07
214 0.6DL + 1.0WLZ2	-169.97	0.92	-1.79	-97.47	-190.07

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	659.9	-1.33	-2.6	-9.7	18.8	78.8	509.4	27.3	198.6
202 1.0DL + 1.0LL	923.4	-3.51	-6.8	-40.1	78.2	26.6	269.7	9.2	105.2
203 1.0DL + 1.0WLX1	787.9	-5.34	-10.4	-76.5	149.2	11.9	151.4	4.1	59.1
204 1.0DL + 1.0WLX2	531.9	2.68	5.2	57.2	-111.5	10.7	203.7	3.7	79.4
205 1.0DL + 1.0WLZ1	887.5	-98.00	-191.1	-16.4	31.9	62.5	9.3	21.7	3.6
206 1.0DL + 1.0WLZ2	432.3	-98.00	-191.1	-2.9	5.7	169.3	4.5	58.7	1.8
207 1.0DL + 0.75WLX1 + 0.75LL	953.6	-5.97	-11.6	-82.6	161.1	13.3	163.8	4.6	63.9
208 1.0DL + 0.75WLX2 + 0.75LL	761.5	0.04	0.1	17.7	-34.4	49.8	20040.6	17.3	7810.7
209 1.0DL + 0.75WLZ1 + 0.75LL	1028.2	-75.47	-147.2	-37.5	73.2	31.6	14.0	11.0	5.4
210 1.0DL + 0.75WLZ2 + 0.75LL	686.9	-75.47	-147.2	-27.5	53.5	28.9	9.3	10.0	3.6
211 0.6DL + 1.0WLX1	749.5	-4.80	-9.4	-72.7	141.7	11.9	160.0	4.1	62.4
212 0.6DL + 1.0WLX2	493.4	3.21	6.3	61.1	-119.1	9.3	157.7	3.2	61.5
213 0.6DL + 1.0WLZ1	849.0	-97.47	-190.1	-12.5	24.4	78.3	8.9	27.1	3.5
214 0.6DL + 1.0WLZ2	393.9	-97.47	-190.1	0.9	-1.8	494.3	4.1	171.3	1.6

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	80.59	0
Min Px	2	201	0	80.59	0
Max Py	634	209	0	170.96	0
Min Py	863	213	0	1.82	0
Max Pz	2	201	0	80.59	0
Min Pz	2	201	0	80.59	0

Gross Maximum Base pressure, p_{max} = 170.96 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.5 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 360 KN/m^2
> 171.0 KN/m^2
Hence OK

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Design of foundation raft F2

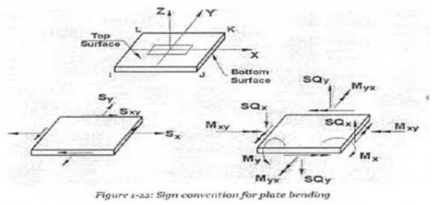
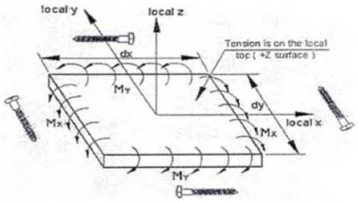


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	427	0.499	0.168	0	0	0	-167.1	-89.969	-59.754	-226.89	-149.72	103:1.4DL + 1.6WLX1 + 1.0 LL
Min Qx	344	-0.375	-0.27	0	0	0	-72.19	-97.569	-46.686	-118.88	-144.26	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Qy	417	0.253	0.34	0	0	0	-109	-70.703	-54.192	-163.23	-124.9	103:1.4DL + 1.6WLX1 + 1.0 LL
Min Qy	346	0.11	-0.613	0	0	0	-116.3	-177.95	-35.534	-151.86	-213.48	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Sx	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sx	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Sy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Sxy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sxy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Mx	427	-0.201	-0.077	0	0	0	78.379	20.669	29.983	108.362	50.652	108:0.9DL + 1.6WLX2
Min Mx	333	0.494	-0.122	0	0	0	-167.3	-94.327	-32.575	-199.85	-126.9	103:1.4DL + 1.6WLX1 + 1.0 LL
Max My	416	-0.118	-0.38	0	0	0	51.638	156.913	25.256	76.894	182.169	110:0.9DL + 1.6WLZ2
Min My	346	0.11	-0.613	0	0	0	-116.3	-177.95	-35.534	-151.86	-213.48	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Mxy	347	0.102	-0.006	0	0	0	-0.705	-6.184	43.033	-43.738	-49.217	110:0.9DL + 1.6WLZ2
Min Mxy	414	-0.013	0.047	0	0	0	-23.51	15.416	-116.59	-140.11	132.008	105:1.4DL + 1.6WLZ1 + 1.0 LL

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	799	0.284	0.002	0	0	0	-82.74	-1.247	0.554	-83.289	-1.801	105:1.4DL + 1.6WLZ1 + 1.0 LL
Min Qx	617	-0.15	-0.027	0	0	0	-15.6	-1.945	-9.596	-25.2	-11.541	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Qy	693	0.084	0.184	0	0	0	-17.5	-34.535	-24.992	-42.496	-59.527	105:1.4DL + 1.6WLZ1 + 1.0 LL
Min Qy	856	0.001	-0.167	0	0	0	0.058	-9.406	-17.196	17.254	-26.602	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max Sx	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sx	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Sy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Sxy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Min Sxy	871	0.001	-0.001	0	0	0	-0.248	-0.235	-0.33	-0.578	-0.565	101:1.4DL
Max Mx	799	0.087	0.016	0	0	0	46.27	0.283	3.753	50.023	4.036	110:0.9DL + 1.6WLZ2
Min Mx	798	-0.1	-0.022	0	0	0	-106.3	-0.165	7.533	-113.8	-7.698	105:1.4DL + 1.6WLZ1 + 1.0 LL
Max My	693	-0.026	0.014	0	0	0	1.334	45.725	6.107	7.441	51.832	110:0.9DL + 1.6WLZ2
Min My	693	0.09	0.15	0	0	0	-17.76	-63.502	-29.344	-47.105	-92.846	103:1.4DL + 1.6WLX1 + 1.0 LL
Max Mxy	692	0.066	0.03	0	0	0	41.126	9.097	19.861	60.987	28.958	110:0.9DL + 1.6WLZ2
Min Mxy	411	-0.016	0.002	0	0	0	-8.469	-7.235	-57.192	-65.661	-64.427	105:1.4DL + 1.6WLZ1 + 1.0 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

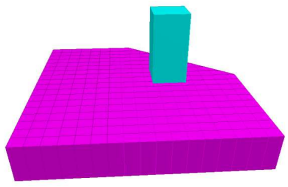
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:0.9D	108.36	top face	50	692	108.36	0.30	0.0007	499.7	1245.6	16	161.48	150	1341	18.9	23.64	0.085	tension control
103:1.4D	-199.85	bottom face	50	692	199.85	0.56	0.0013	925.6		16	161.48	150	1341	18.9	23.64	0.085	tension control
110:0.9D	-43.74	bottom face	50	692	43.738	0.12	0.0003	201.1		16	161.48	150	1341	18.9	23.64	0.085	tension control
105:1.4D	-140.11	bottom face	50	692	140.11	0.39	0.0009	647.0		16	161.48	150	1341	18.9	23.64	0.085	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.284	1	213	83.289	0.002	0.692	695.372	677.825	1187.24	677.83	2535	> 213	ok	508.4	φVc>Vu	not required
	Min Qx	0.15	1	112.5	25.2			725.455			677.83	2535	> 112.5	ok	508.4	φVc>Vu	not required

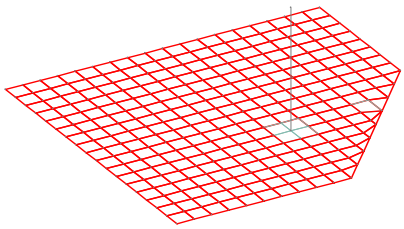
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
110:0.9D	182.17	top face	50	692	182.17	0.51	0.0012	843.0	1245.6	16	161.48	150	1341	18.9	23.64	0.085	tension control
105:1.4D	-213.48	bottom face	50	692	213.48	0.59	0.0014	989.3		16	161.48	150	1341	18.9	23.64	0.085	tension control
110:0.9D	-49.22	bottom face	50	692	49.217	0.14	0.0003	226.3		16	161.48	150	1341	18.9	23.64	0.085	tension control
105:1.4D	132.01	top face	50	692	132.01	0.37	0.0009	609.4		16	161.48	150	1341	18.9	23.64	0.085	tension control

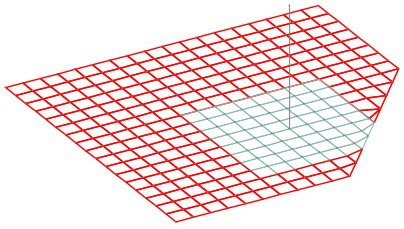
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.184	1	138	59.527	0.002	0.692	691.6	677.825	fc'1187	677.83	711	> 138	ok	508.4	φVc>Vu	not required
	Min Qy	0.167	1	125.25	26.602			729.304			677.83	711	> 125.25	ok	508.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **16ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-X	Along-Z	
2	0.50m	0.50m	
311	0.50m	0.50m	
Size of fdn. raft:	2.275m	1.00m	
Thickness of raft:	0.50	m	
Original Gr. Level:	0.00	m	
Finished Gr. Level:	0.00	m	
U/S of B/Pl Above GL:	0.70	m	
U/S level of fdn:	-1.25	m	
Unit weight of conc:	25	kN/m^3	
Unit weight of soil:	18	kN/m^3	

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

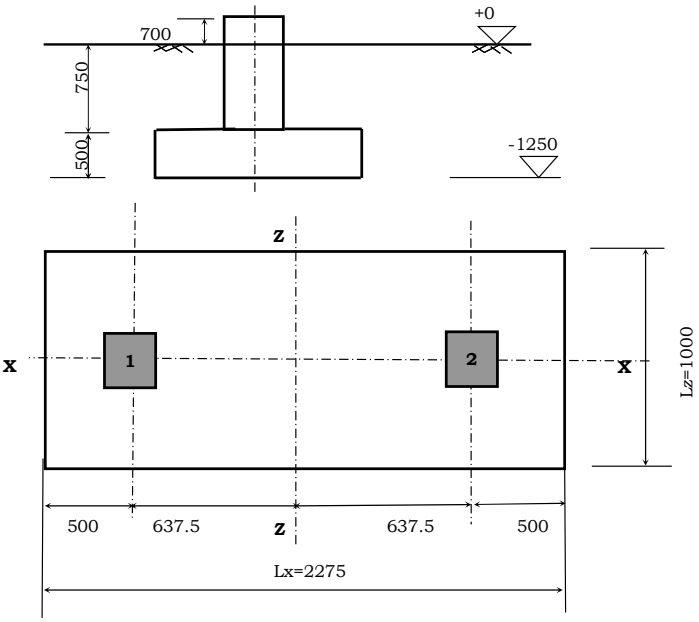
Calculation of Weight of Foundation :

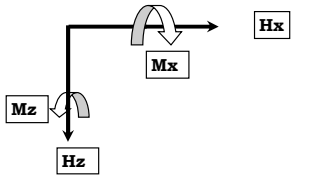
Weight of raft = (2.275*1*0.5*25) = 28.44KN

Weight of soil = ((2.275*1)-2*(0.5*0.5))*0.75*18 = 23.96KN

Total = 52.40KN

23.96KN





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of fooring	Node 2					Node 311				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	15.69	0.13	-0.26	0.12	0.24	15.92	-0.13	0.26	0.04	0.08
202 1.0DL + 1.0LL	26.51	0.56	-1.09	0.62	1.21	27.05	-0.56	1.09	0.21	0.41
203 1.0DL + 1.0WLX1	13.25	-3.83	7.47	0.12	0.23	18.37	-1.48	2.89	0.04	0.08
204 1.0DL + 1.0WLX2	18.14	4.09	-7.98	0.12	0.24	13.48	1.22	-2.38	0.04	0.08
205 1.0DL + 1.0WLZ1	15.24	-0.07	0.13	-0.67	-1.30	15.70	-0.19	0.38	-0.56	-1.10
206 1.0DL + 1.0WLZ2	16.15	0.33	-0.64	0.91	1.77	16.14	-0.07	0.13	0.64	1.26
207 1.0DL + 0.75WLX1 + 0.75LL	21.98	-2.52	4.91	0.50	0.96	26.10	-1.47	2.86	0.17	0.33
208 1.0DL + 0.75WLX2 + 0.75LL	25.64	3.43	-6.68	0.50	0.97	22.43	0.56	-1.09	0.17	0.33
209 1.0DL + 0.75WLZ1 + 0.75LL	23.47	0.30	-0.59	-0.09	-0.18	24.10	-0.50	0.98	-0.29	-0.56
210 1.0DL + 0.75WLZ2 + 0.75LL	24.15	0.60	-1.18	1.09	2.12	24.43	-0.41	0.79	0.62	1.21
211 0.6DL + 1.0WLX1	6.97	-3.88	7.57	0.07	0.14	12.00	-1.43	2.78	0.03	0.05
212 0.6DL + 1.0WLX2	11.86	4.04	-7.88	0.08	0.15	7.11	1.27	-2.48	0.03	0.05
213 0.6DL + 1.0WLZ1	8.96	-0.12	0.24	-0.71	-1.39	9.34	-0.14	0.28	-0.58	-1.13
214 0.6DL + 1.0WLZ2	9.87	0.28	-0.54	0.86	1.67	9.77	-0.02	0.03	0.63	1.22

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	84.0	0.00	0.1	0.2	0.3	657.5	132.9	NA	207.4
202 1.0DL + 1.0LL	106.0	0.00	0.3	0.8	1.6	355.4	32.7	NA	51.0
203 1.0DL + 1.0WLX1	84.0	-5.31	13.6	0.2	0.3	7.0	135.5	6.3	211.4
204 1.0DL + 1.0WLX2	84.0	5.31	-13.3	0.2	0.3	7.2	130.5	6.3	203.7
205 1.0DL + 1.0WLZ1	83.3	-0.26	0.8	-1.2	-2.4	117.4	17.4	127.2	27.2
206 1.0DL + 1.0WLZ2	84.7	0.26	-0.5	1.6	3.0	186.8	14.0	129.3	21.8
207 1.0DL + 0.75WLX1 + 0.75LL	100.5	-3.98	10.4	0.7	1.3	11.0	38.9	10.1	60.7
208 1.0DL + 0.75WLX2 + 0.75LL	100.5	3.98	-9.8	0.7	1.3	11.6	38.7	10.1	60.3
209 1.0DL + 0.75WLZ1 + 0.75LL	100.0	-0.20	0.8	-0.4	-0.7	144.5	67.8	203.0	105.8
210 1.0DL + 0.75WLZ2 + 0.75LL	101.0	0.20	-0.2	1.7	3.3	558.2	15.2	205.0	23.7
211 0.6DL + 1.0WLX1	71.4	-5.31	13.6	0.1	0.2	6.0	193.9	5.4	300.5
212 0.6DL + 1.0WLX2	71.4	5.31	-13.4	0.1	0.2	6.1	183.0	5.4	282.6
213 0.6DL + 1.0WLZ1	70.7	-0.26	0.7	-1.3	-2.5	107.3	14.0	107.9	21.9
214 0.6DL + 1.0WLZ2	72.0	0.26	-0.6	1.5	2.9	142.6	12.4	110.0	19.4

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	61.38	0
Min Px	2	201	0	61.38	0
Max Py	477	207	0	83.26	0
Min Py	322	211	0	20.91	0
Max Pz	2	201	0	61.38	0
Min Pz	2	201	0	61.38	0

Gross Maximum Base pressure, p_{max} = 83.3 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.75 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 365 KN/m^2
> 83.3 KN/m^2
Hence OK

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Design of foundation raft F3

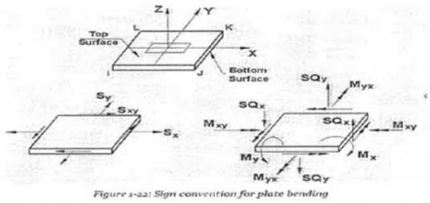
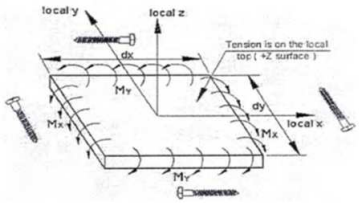


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	427	0.043	0.008	0	0	0	-4.998	-1.707	-0.455	-5.453	-2.162	103:1.4DL + 1.6WLX1 + 1.0 LL
Min Qx	337	-0.036	-0.001	0	0	0	4.029	-1.496	0	4.029	-1.496	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Qy	414	-0.004	0.02	0	0	0	-0.359	-1.108	0.274	-0.633	-1.382	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Qy	344	-0.002	-0.018	0	0	0	-0.251	-1.03	-0.198	-0.449	-1.228	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Sx	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sx	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Sy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Sxy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sxy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Mx	333	-0.007	0.006	0	0	0	9.875	-1.565	-1.154	11.029	-2.719	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mx	427	0.033	0.007	0	0	0	-6.663	-0.87	-0.434	-7.097	-1.304	107:0.9DL + 1.6WLX1
Max My	345	-0.015	0	0	0	0	-3.836	0.266	0.182	-4.018	0.448	107:0.9DL + 1.6WLX1
Min My	327	-0.024	-0.012	0	0	0	-1.322	-2.877	-0.312	-1.634	-3.189	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Mxy	417	0.003	-0.005	0	0	0	7.768	-0.103	0.897	8.665	-1	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mxy	347	-0.005	0.007	0	0	0	7.892	-0.006	-1.347	9.239	-1.353	104:1.4DL + 1.6WLX2 + 1.0 LL

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	427	0.043	0.008	0	0	0	-4.998	-1.707	-0.455	-5.453	-2.162	103:1.4DL + 1.6WLX1 + 1.0 LL
Min Qx	337	-0.036	-0.001	0	0	0	4.029	-1.496	0	4.029	-1.496	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Qy	414	-0.004	0.02	0	0	0	-0.359	-1.108	0.274	-0.633	-1.382	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Qy	344	-0.002	-0.018	0	0	0	-0.251	-1.03	-0.198	-0.449	-1.228	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Sx	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sx	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Sy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Sxy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Min Sxy	486	0.007	0.002	0	0	0	-0.256	-1.401	0.146	-0.402	-1.547	101:1.4DL
Max Mx	333	-0.007	0.006	0	0	0	9.875	-1.565	-1.154	11.029	-2.719	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mx	427	0.033	0.007	0	0	0	-6.663	-0.87	-0.434	-7.097	-1.304	107:0.9DL + 1.6WLX1
Max My	345	-0.015	0	0	0	0	-3.836	0.266	0.182	-4.018	0.448	107:0.9DL + 1.6WLX1
Min My	327	-0.024	-0.012	0	0	0	-1.322	-2.877	-0.312	-1.634	-3.189	104:1.4DL + 1.6WLX2 + 1.0 LL
Max Mxy	417	0.003	-0.005	0	0	0	7.768	-0.103	0.897	8.665	-1	104:1.4DL + 1.6WLX2 + 1.0 LL
Min Mxy	347	-0.005	0.007	0	0	0	7.892	-0.006	-1.347	9.239	-1.353	104:1.4DL + 1.6WLX2 + 1.0 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

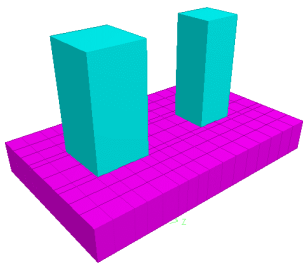
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.4D	11.03	top face	50	442	11.029	0.08	0.0002	79.3	795.6	16	252.82	200	1005.8	14.2	17.73	0.072	tension control
107:0.9D	-7.10	bottom face	50	442	7.097	0.05	0.0001	51.0		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
104:1.4D	8.67	top face	50	442	8.665	0.06	0.0001	62.3		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
104:1.4D	9.24	top face	50	442	9.239	0.06	0.0002	66.4		16	252.82	200	1005.8	14.2	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.043	1	21.5	5.453	0.002	0.442	448.183	435.484	758.323	435.48	1621	> 21.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.036	1	18	4.029			452.149			435.48	1621	> 18	ok	326.6	φVc>Vu	not required

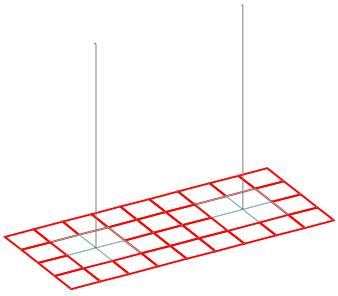
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:0.9D	0.45	top face	50	442	0.448	0.00	0.0000	3.2	795.6	16	252.82	200	1005.8	14.2	17.73	0.072	tension control
104:1.4D	-3.19	bottom face	50	442	3.189	0.02	0.0001	22.9		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
104:1.4D	-1.00	bottom face	50	442	1	0.01	0.0000	7.2		16	252.82	200	1005.8	14.2	17.73	0.072	tension control
104:1.4D	-1.35	bottom face	50	442	1.353	0.01	0.0000	9.7		16	252.82	200	1005.8	14.2	17.73	0.072	tension control

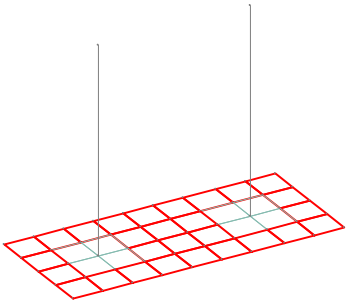
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	104:1.4D	0.02	1	10	1.382	0.002	0.442	473.071	435.484	fc'758.3	435.48	456	> 10	ok	326.6	φVc>Vu	not required
	104:1.4D	0.018	1	9	1.228			473.775			435.48	456	> 9	ok	326.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**16ϕT**	**200**	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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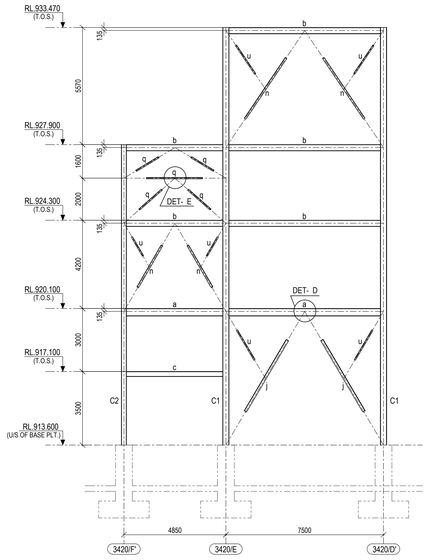
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ATTACHMENT 1 INPUT DRAWINGS

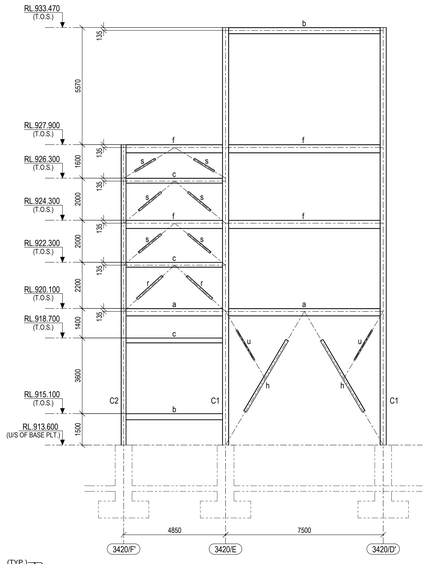
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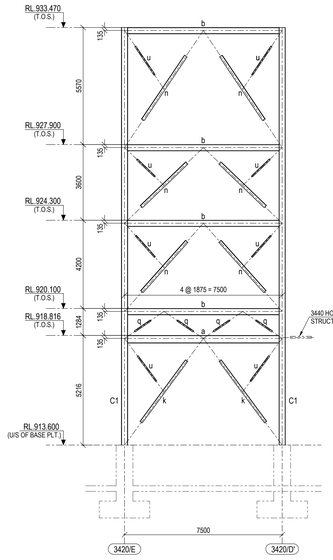
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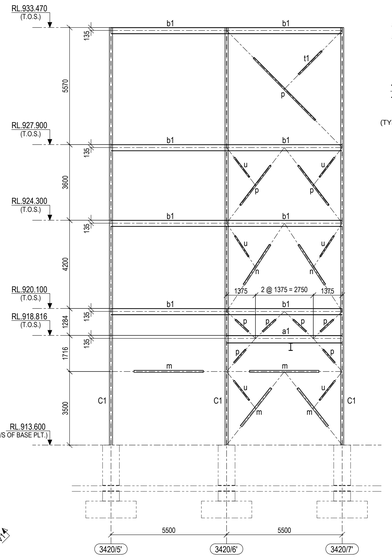
ELEVATION ON GRID- 3420/5'
(SCALE:- 1:100)



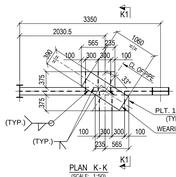
ELEVATION ON GRID- 3420/6'
(SCALE:- 1:100)



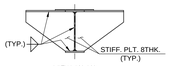
ELEVATION ON GRID- 3420/7'
(SCALE:- 1:100)



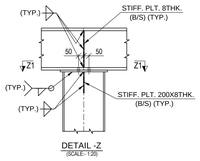
CROSS SECTION AT LINE- 3420/E
(SCALE:-1:100)



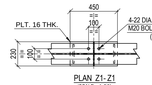
PLAN K-K
(SCALE: 1:50)



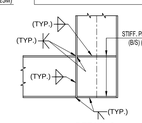
VIEW K1-K1
(SCALE: 1:20)



DETAIL -Z
(SCALE:-1:20)

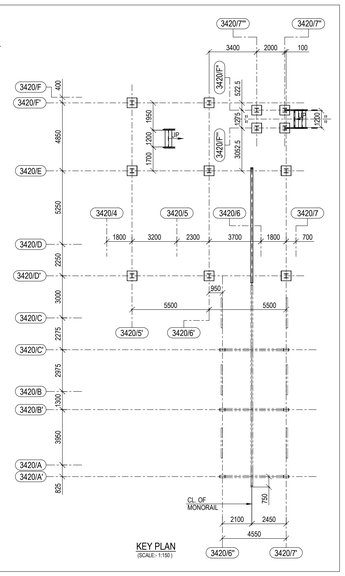


PLAN Z1- Z1
(SCALE: 1:50)

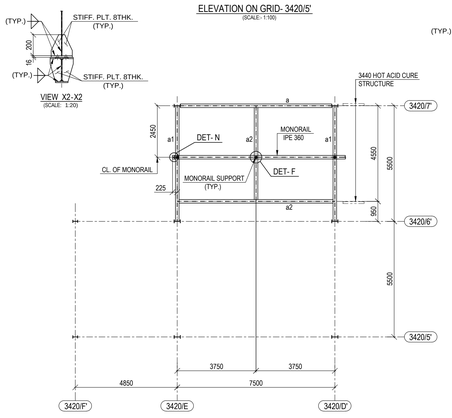


DETAIL -Z
(SCALE:-1:20)



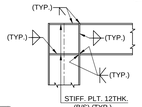


KEY PLAN
(SCALE:-1:150)

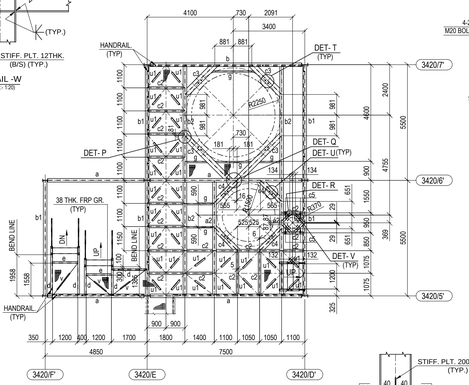


VIEW X2-X2
(SCALE: 1:20)

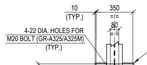
PLAN AT RL 918.816
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE:- 1:100)



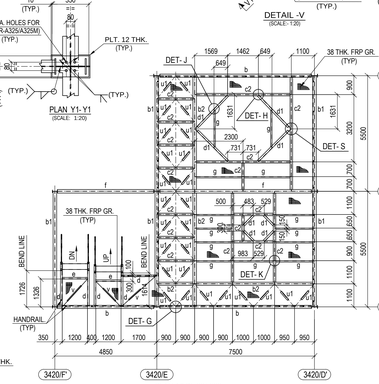
DETAIL -W
(SCALE:-1:20)



PLAN AT RL 920.100
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE:- 1:100)



PLAN Y1- Y1
(SCALE: 1:20)



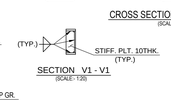
PLAN AT RL 924.300
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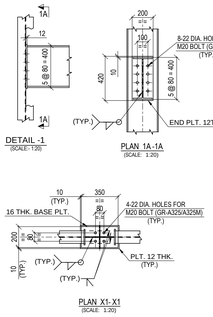
DETAIL -Y
(SCALE:-1:20)



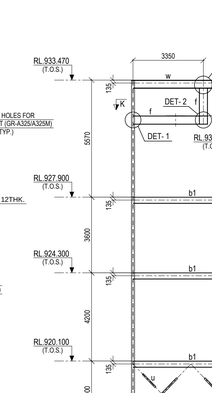
DETAIL -V
(SCALE:-1:20)



SECTION V1 - V1
(SCALE:-1:20)



PLAN X1- X1
(SCALE: 1:20)



CROSS SECTION AT LINE- 3420/D'
(SCALE:-1:100)

DETAIL -Z
(SCALE:-1:20)

PLAN AT RL 913.600
(SCALE:- 1:100)

VIEW J - J
(SCALE: 1:100)

PLAN AT RL 927.900
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE:- 1:100)



SECTION T1 - T1
(SCALE:-1:20)

PLAN AT RL 933.470
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE:- 1:100)

SECTION J2 - J2
(SCALE:-1:20)

PLAN J1- J1
(SCALE: 1:50)

CROSS SECTION AT LINE- 3420/F'
(SCALE:-1:100)

SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
C1	IPE 300	I	-	-	
C2	IPE 240	I	-	-	
a	IPE360	I	17.0	23.0	
a1	IPE360	I	5.0	7.0	
a2	IPE360	I	8.0	11.0	
b	IPE300	I	8.0	8.0	
b1	IPE300	I	21.0	7.0	
b2	IPE300	I	7.0	8.0	
b3	1/WEB PLT. 268x10 THK. 2/FLG. PLT. 200x12 THK.	I	21.0	7.0	MATERIAL GRADE:S275 JR
c	IPE240	I	5.0	5.0	
c1	IPE240	I	-	-	
c2	IPE240	I	5.0	9.0	
c3	1/WEB PLT. 208x10 THK. 2/FLG. PLT. 300x16 THK.	I	4.0	4.0	
c4	1/WEB PLT. 208x10 THK. 2/FLG. PLT. 250x16 THK.	I	6.0	5.0	
c5	1/WEB PLT. 208x10 THK. 2/FLG. PLT. 90x16 THK.	I	4.0	-	
d	UPN240	C	2.0	5.0	
d1	UPN240	C	-	-	
e	UPN200	C	-	-	
f	IPE400	I	6.0	14.0	
f1	IPE400	I	3.0	-	
g	UPN140	C	3.0	-	
h	L - 150X150X13	⊣⊢	39.0	-	
j	L - 130X130X13	⊣⊢	27.0	-	
k	L - 130X130X10	⊣⊢	22.0	-	
m	L - 100X100X8	⊣⊢	27.0	-	
n	L - 90X90X8	⊣⊢	26.0	-	
p	L - 75X75X8	⊣⊢	29.0	-	
q	L - 65X65X6	⊣⊢	11.0	-	
r	L - 100X100X8	⌐	7.0	-	
s	L - 90X90X8	⌐	6.0	-	
t	L - 75X75X8	⌐	-	-	
t1	L - 75X75X8	⌐	-	-	
u	L - 65X65X6	⌐	-	-	
u1	L - 65X65X6	⌐	6.0	-	
v	L - 50X50X6	⌐	-	-	
w	1/WEB PLT. 350x10 THK. 2/FLG. PLT. 200x25 THK.	I	-	14.0	

ALL GUSSET PLATE ARE 10mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. HANDRAIL PIPE / POST MATERIAL SHALL BE AS PER ASTM A53 Gr.B
4. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
5. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
6. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
7. THIS DRAWING READ IN CONJUCTION WITH DRAWING NO. MD-216-3420-EG-ST-GAD-2239.

LEGENDS:-
TOS - TOP OF STEEL
U/S - UNDER SIDE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN - DOWN
W.P. WORKING POINT
GR. - GRATING
MC - MOMENT CONNECTION
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS

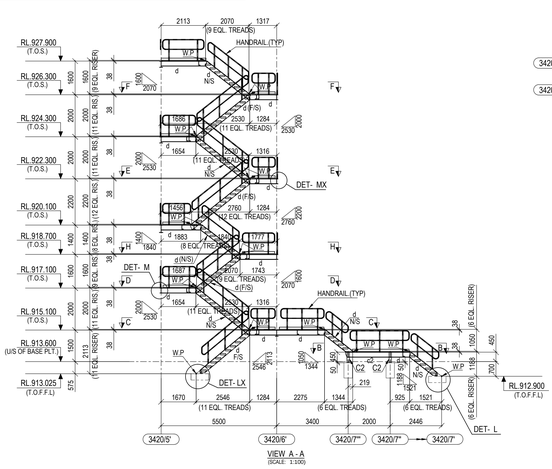
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MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	07/05/2020	ISSUED FOR APPROVAL	MK	SYC	SGI	PRI	
		A02	12/06/2020	REVISED AS PER EMPLOYER'S COMMENTS DATED 09/06/2020	MK	SYC	SGI	PRI	

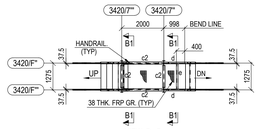
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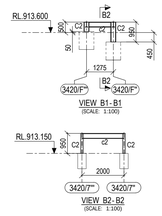
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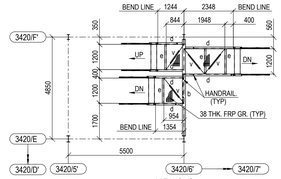
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OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
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CHECKED	SYC	04/05/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SGI	05/05/2020	PRESSURE OXIDATION SCRUBBER SUPPORT
APPROVED	PRI	07/05/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1 : 20, 50, 100 & 150	A0	DRAWING NUMBER: MD-216-3420-EG-ST-GAD-2238 REV: A02

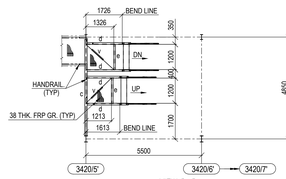
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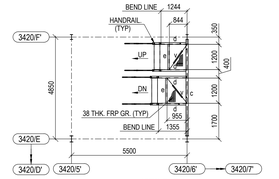


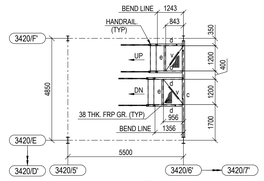


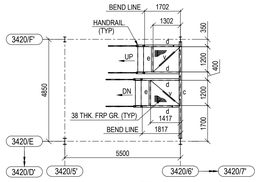


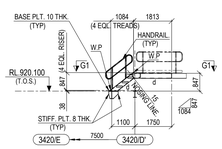


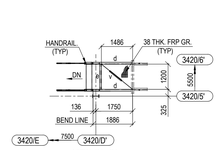


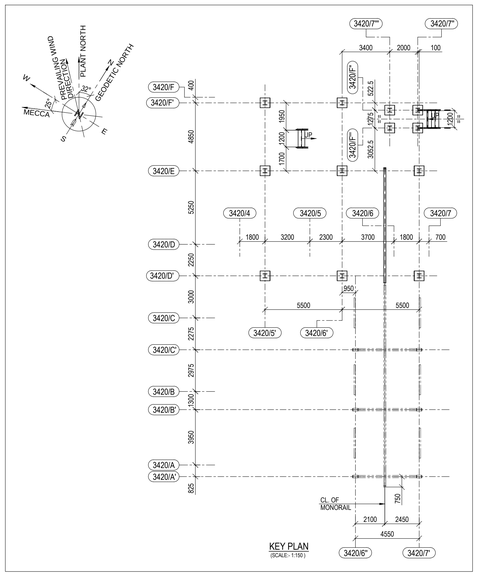


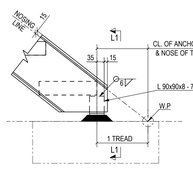


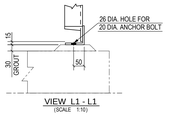


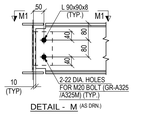


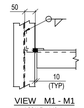


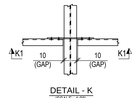


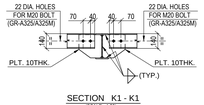


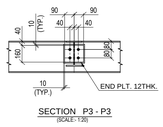


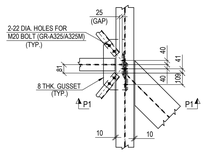


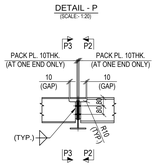


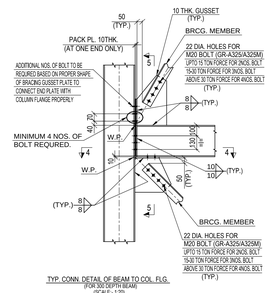


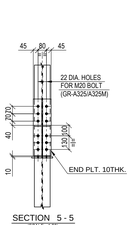


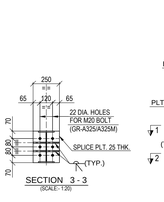


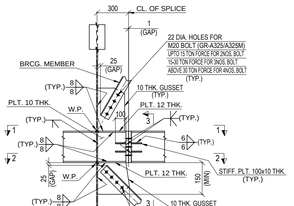


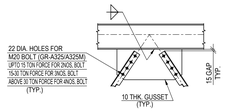


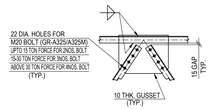


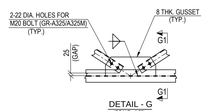




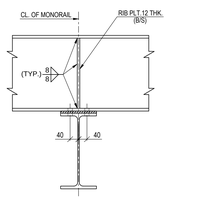


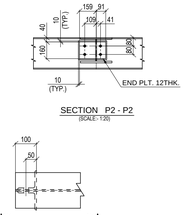


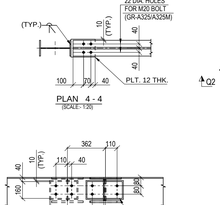


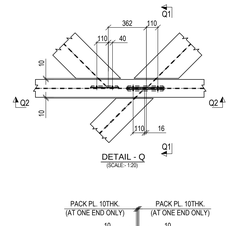


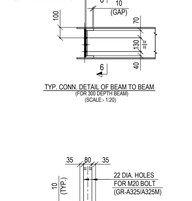


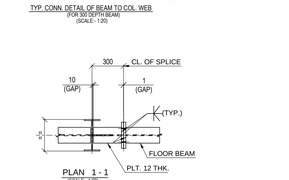


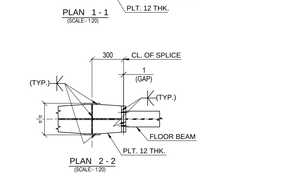


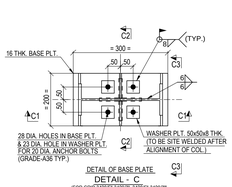


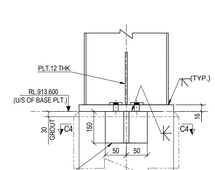


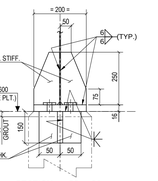


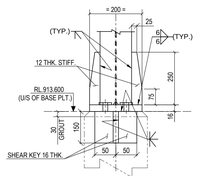


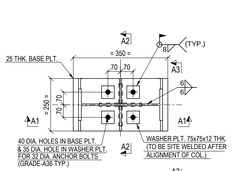


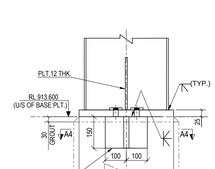


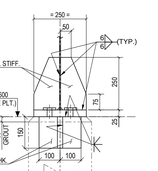


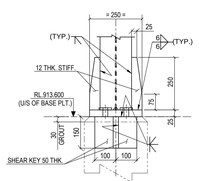


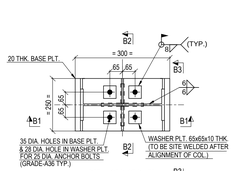


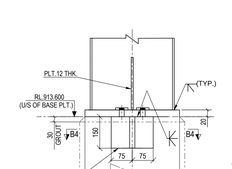


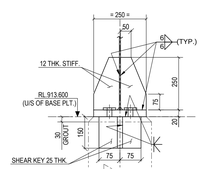


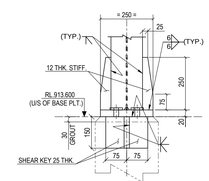


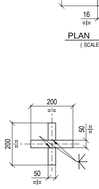


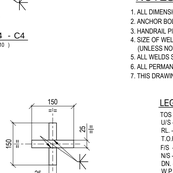


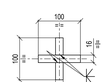












NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. HANDRAIL PIPE / POST MATERIAL SHALL BE AS PER ASTM A53 Gr.B
4. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
5. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
6. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
7. THIS DRAWING READ IN CONJUCTION WITH DRAWING NO. MD-216-3420-EG-ST-GAD-2238.

LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN - DOWN
W.P. - WORKING POINT
GR. - GRATING
MC - MOMENT CONNECTION
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	12/06/2020	REVISED AS PER EMPLOYER'S COMMENTS DATED 09/06/2020	MK	SYC	SGI	PRI	
A01	07/05/2020	ISSUED FOR APPROVAL	MK	SYC	SGI	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	07/05/2020	ISSUED FOR APPROVAL	MK	SYC	SGI	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 0190L5_450_S_G_26043_011 — REVISION: 1

OUTOTEC DRAWING NUMBER: 900707__GAU01__HH12__MTL01__34221 — DOCUMENT ID: OU100668484 — REVISION: 1

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DRAWN	MK	02/05/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	04/05/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SGI	05/05/2020	PRESSURE OXIDATION SCRUBBER SUPPORT
APPROVED	PRI	07/05/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1 : 10, 20, 100 & 150	A0	DRAWING NUMBER: MD-216-3420-EG-ST-GAD-2239 — REV: A02

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175;
35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175;
128 0 913.6 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438;
304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438;
307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175;
311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438;
315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438;
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323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438;
326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438;
329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438;
332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438;
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368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925;
371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438;
374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925;
377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438;
380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925;
383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925;
386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438;
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392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438;
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398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438;
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404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438;
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416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438;
419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175;
422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438;
425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925;
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431 -1.5 909.9 3.42438; 432 -1.5 909.9 3.175; 433 -1.5 909.9 3.67438;
434 -1.5 909.9 3.92438; 435 -1.5 909.9 4.17438; 436 -1.5 909.9 4.42438;
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440 -1.5 909.9 2.67438; 441 -1.5 909.9 2.92438; 442 -0.75 909.9 4.67438;
443 -0.5 909.9 4.67438; 444 -0.25 909.9 4.67438; 445 0 909.9 4.67438;
446 0.25 909.9 4.67438; 447 0.5 909.9 4.67438; 448 0.75 909.9 4.67438;
449 1 909.9 4.67438; 450 1.25 909.9 4.67438; 451 -1.25 909.9 4.67438;
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455 -0.5 909.9 1.675; 456 -0.75 909.9 1.675; 457 -0.25 909.9 1.675;
458 0 909.9 1.675; 459 0.25 909.9 1.675; 460 0.5 909.9 1.675;
461 0.75 909.9 1.675; 462 1 909.9 1.675; 463 1.25 909.9 1.675;
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479 1.75 909.9 4.67438; 480 1.75 909.9 1.675; 481 -1.75 909.9 3.42438;
482 -1.75 909.9 3.175; 483 -1.75 909.9 3.67438; 484 -1.75 909.9 3.92438;
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494 -0.5 909.9 1.425; 495 -0.75 909.9 1.425; 496 -0.25 909.9 1.425;
497 0 909.9 1.425; 498 0.25 909.9 1.425; 499 0.5 909.9 1.425;
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497 0 909.9 1.425; 498 0.25 909.9 1.425; 499 0.5 909.9 1.425;
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503 -1 909.9 1.425; 504 -1.25 909.9 1.425; 505 1.5 909.9 1.425;
506 -1.5 909.9 1.425; 507 1.75 909.9 1.425; 508 -1.75 909.9 1.425;
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515 0.75 909.9 4.92438; 516 1 909.9 4.92438; 517 1.25 909.9 4.92438;
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521 -1.5 909.9 4.92438; 522 1.75 909.9 4.92438; 523 -1.75 909.9 4.92438;
524 2 909.9 3.42438; 525 2 909.9 3.175; 526 2 909.9 3.67438;
527 2 909.9 3.92438; 528 2 909.9 4.17438; 529 2 909.9 4.42438;
530 2 909.9 2.17438; 531 2 909.9 1.925; 532 2 909.9 2.42438;
533 2 909.9 2.67438; 534 2 909.9 2.92438; 535 2 909.9 4.67438;
536 2 909.9 1.675; 537 2 909.9 1.425; 538 2 909.9 4.92438;
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542 -2 909.9 3.92438; 543 -2 909.9 4.17438; 544 -2 909.9 4.42438;
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563 -1 909.9 1.175; 564 -1.25 909.9 1.175; 565 1.5 909.9 1.175;
566 -1.5 909.9 1.175; 567 1.75 909.9 1.175; 568 -1.75 909.9 1.175;
569 2 909.9 1.175; 570 -2 909.9 1.175; 571 -0.75 909.9 5.17438;
572 -0.5 909.9 5.17438; 573 -0.25 909.9 5.17438; 574 0 909.9 5.17438;
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581 -1 909.9 5.17438; 582 1.5 909.9 5.17438; 583 -1.5 909.9 5.17438;
584 1.75 909.9 5.17438; 585 -1.75 909.9 5.17438; 586 2 909.9 5.17438;
587 -2 909.9 5.17438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309;
341 319 320 316 315; 342 320 303 10 316; 343 321 322 302 303;
344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305;
347 325 326 307 306; 348 326 327 308 307; 349 327 328 313 308;
350 328 329 314 313; 351 330 331 320 319; 352 331 321 303 320;
353 332 333 322 321; 354 333 334 323 322; 355 334 335 324 323;
356 335 336 325 324; 357 336 337 326 325; 358 337 338 327 326;
359 338 339 328 327; 360 339 340 329 328; 361 341 342 331 330;
362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333;
365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336;
368 348 349 338 337; 369 349 350 339 338; 370 350 351 340 339;
371 352 353 342 341; 372 353 343 332 342; 373 354 355 344 343;
374 355 356 345 344; 375 356 357 346 345; 376 357 358 347 346;
377 358 359 348 347; 378 359 360 349 348; 379 360 361 350 349;
380 361 362 351 350; 381 363 364 353 352; 382 364 354 343 353;
383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370;
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389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386;
392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366;
395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373;
398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379;
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413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
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422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415;
430 309 311 417 416; 431 315 316 419 418; 432 316 10 409 419;
433 314 420 421 311; 434 329 422 420 314; 435 340 423 422 329;
436 351 424 423 340; 437 362 425 424 351; 438 381 426 427 382;
439 395 428 426 381; 440 406 429 428 395; 441 417 430 429 406;
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445 434 341 330 433; 446 435 352 341 434; 447 436 363 352 435;
448 437 383 386 438; 449 439 396 383 437; 450 440 407 396 439;
451 441 418 407 440; 452 432 315 418 441; 453 442 443 355 354;
454 443 444 356 355; 455 444 445 357 356; 456 445 446 358 357;
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466 367 370 457 455; 467 370 372 458 457; 468 372 374 459 458;
469 374 376 460 459; 470 376 378 461 460; 471 378 380 462 461;
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469 374 376 460 459; 470 376 378 461 460; 471 378 380 462 461;
472 380 382 463 462; 473 386 385 464 465; 474 385 368 456 464;
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478 422 470 468 420; 479 423 471 470 422; 480 424 472 471 423;
481 425 473 472 424; 482 426 474 475 427; 483 428 476 474 426;
484 429 477 476 428; 485 430 478 477 429; 486 421 469 478 430;
487 453 479 473 425; 488 427 475 480 466; 489 481 431 432 482;
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496 490 440 439 489; 497 491 441 440 490; 498 482 432 441 491;
499 492 454 436 486; 500 488 438 467 493; 501 456 455 494 495;
502 455 457 496 494; 503 457 458 497 496; 504 458 459 498 497;
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529 468 524 525 469; 530 470 526 524 468; 531 471 527 526 470;
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544 541 483 481 539; 545 542 484 483 541; 546 543 485 484 542;
547 544 486 485 543; 548 545 487 488 546; 549 547 489 487 545;
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553 550 492 486 544; 554 546 488 493 551; 555 551 493 508 552;
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559 496 497 557 556; 560 497 498 558 557; 561 498 499 559 558;
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583 579 582 520 517; 584 583 580 518 521; 585 582 584 522 520;
586 585 583 521 523; 587 584 586 538 522; 588 587 585 523 553;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 338 341 TO 588
_RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 588
_RAFT_SHEAR 338 341 350 351 360 361 370 371 373 TO 392 400 401 410 411 420 -
421 430 431 433 TO 588
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 338 341 TO 588 THICKNESS 0.75
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 338 341 TO 587 -
588 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -47
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -20
JOINT LOAD
128 FX -25.516 FY -145.216 FZ 0.208
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
```



```
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX -58.516 FY -379.137 FZ -1.923
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 89.549 FY 82.052 FZ -4.011
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX -89.549 FY -82.052 FZ 4.011
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX -16.504 FY -361.777 FZ 103.166
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 16.504 FY 361.777 FZ -103.166
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL + 1.7LL
1 1.4 2 1.7
LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
1 1.4 3 1.6 2 1.0
LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
1 1.4 4 1.6 2 1.0
LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
1 1.4 5 1.6 2 1.0
LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
1 1.4 6 1.6 2 1.0
LOAD COMB 107 0.9DL + 1.6WLX1
1 0.9 3 1.6
LOAD COMB 108 0.9DL + 1.6WLX2
1 0.9 4 1.6
LOAD COMB 109 0.9DL + 1.6WLZ1
1 0.9 5 1.6
LOAD COMB 110 0.9DL + 1.6WLZ2
1 0.9 6 1.6
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
```

```
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 913.6 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX -25.516 FY -145.216 FZ 0.208
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX -58.516 FY -379.137 FZ -1.923
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 89.549 FY 82.052 FZ -4.011
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX -89.549 FY -82.052 FZ 4.011
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX -16.504 FY -361.777 FZ 103.166
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 16.504 FY 361.777 FZ -103.166
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
```



```
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175;
35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175;
128 0 911.85 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438;
304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438;
307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175;
313 1 909.9 3.42438; 315 -1.25 909.9 3.175; 316 -1 909.9 3.175;
319 -1.25 909.9 3.42438; 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438;
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325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438;
330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438; 365 -0.75 909.9 2.17438;
366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925; 368 -0.75 909.9 1.925;
369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925; 371 0 909.9 2.17438;
372 0 909.9 1.925; 373 0.25 909.9 2.17438; 374 0.25 909.9 1.925;
375 0.5 909.9 2.17438; 376 0.5 909.9 1.925; 377 0.75 909.9 2.17438;
378 0.75 909.9 1.925; 379 1 909.9 2.17438; 380 1 909.9 1.925;
381 1.25 909.9 2.17438; 382 1.25 909.9 1.925; 383 -1.25 909.9 2.17438;
384 -1 909.9 2.17438; 385 -1 909.9 1.925; 386 -1.25 909.9 1.925;
387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438;
390 0 909.9 2.42438; 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438;
393 0.75 909.9 2.42438; 394 1 909.9 2.42438; 395 1.25 909.9 2.42438;
396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438; 398 -0.75 909.9 2.67438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
405 1 909.9 2.67438; 406 1.25 909.9 2.67438; 407 -1.25 909.9 2.67438;
408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438; 410 -0.5 909.9 2.92438;
411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438;
414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438;
417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438; 419 -1 909.9 2.92438;
455 -0.5 909.9 1.675; 456 -0.75 909.9 1.675; 457 -0.25 909.9 1.675;
458 0 909.9 1.675; 459 0.25 909.9 1.675; 460 0.5 909.9 1.675;
461 0.75 909.9 1.675; 462 1 909.9 1.675; 463 1.25 909.9 1.675;
464 -1 909.9 1.675; 465 -1.25 909.9 1.675; 609 -0.75 909.9 4.42438;
610 -0.5 909.9 4.42438; 611 -0.5 909.9 4.175; 612 -0.75 909.9 4.175;
613 -0.25 909.9 4.42438; 614 -0.25 909.9 4.175; 615 0 909.9 4.42438;
616 0 909.9 4.175; 618 0.25 909.9 4.175; 627 -1.25 909.9 4.42438;
628 -1 909.9 4.42438; 629 -1 909.9 4.175; 630 -1.25 909.9 4.175;
631 -0.75 909.9 4.67438; 632 -0.5 909.9 4.67438; 633 -0.25 909.9 4.67438;
634 0 909.9 4.67438; 640 -1.25 909.9 4.67438; 641 -1 909.9 4.67438;
642 -0.75 909.9 3.92438; 643 -0.5 909.9 3.92438; 644 -0.25 909.9 3.92438;
645 0 909.9 3.92438; 646 0.25 909.9 3.92438; 647 0.5 909.9 3.92438;
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690 -0.75 909.9 1.425; 691 -0.25 909.9 1.425; 692 0 909.9 1.425;
693 0.25 909.9 1.425; 694 0.5 909.9 1.425; 695 0.75 909.9 1.425;
696 1 909.9 1.425; 697 1.25 909.9 1.425; 698 -1 909.9 1.425;
699 -1.25 909.9 1.425; 700 -0.5 909.9 1.175; 701 -0.75 909.9 1.175;
702 -0.25 909.9 1.175; 703 0 909.9 1.175; 704 0.25 909.9 1.175;
705 0.5 909.9 1.175; 706 0.75 909.9 1.175; 707 1 909.9 1.175;
708 1.25 909.9 1.175; 709 -1 909.9 1.175; 710 -1.25 909.9 1.175;
711 1.5 909.9 1.425; 712 1.5 909.9 1.175; 713 -1.5 909.9 1.425;
714 -1.5 909.9 1.175; 715 -1.75 909.9 3.42438; 716 -1.75 909.9 3.175;
717 -1.75 909.9 3.67438; 718 -1.75 909.9 2.17438; 719 -1.75 909.9 1.925;
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726 -1.75 909.9 4.67438; 727 -1.75 909.9 3.92438; 728 -2 909.9 3.42438;
729 -2 909.9 3.175; 730 -2 909.9 3.67438; 731 -2 909.9 2.17438;
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735 -2 909.9 2.92438; 736 -2 909.9 1.675; 737 -2 909.9 4.42438;
738 -2 909.9 4.175; 739 -2 909.9 4.67438; 740 -2 909.9 3.92438;
741 -1.75 909.9 1.425; 742 -1.75 909.9 1.175; 743 -2 909.9 1.425;
744 -2 909.9 1.175; 745 1.75 909.9 2.17438; 746 1.75 909.9 1.925;
747 1.75 909.9 2.42438; 750 1.75 909.9 1.675; 751 2 909.9 2.17438;
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747 1.75 909.9 2.42438; 750 1.75 909.9 1.675; 751 2 909.9 2.17438;
752 2 909.9 1.925; 753 2 909.9 2.42438; 756 2 909.9 1.675;
761 1.75 909.9 1.425; 762 1.75 909.9 1.175; 763 2 909.9 1.425;
764 2 909.9 1.175; 765 -0.5 909.9 0.925; 766 -0.75 909.9 0.925;
767 -0.25 909.9 0.925; 768 0 909.9 0.925; 769 0.25 909.9 0.925;
770 0.5 909.9 0.925; 771 0.75 909.9 0.925; 772 1 909.9 0.925;
773 1.25 909.9 0.925; 774 -1 909.9 0.925; 775 -1.25 909.9 0.925;
776 1.5 909.9 0.925; 777 -1.5 909.9 0.925; 778 -1.75 909.9 0.925;
779 -2 909.9 0.925; 780 1.75 909.9 0.925; 781 2 909.9 0.925;
789 0.25 909.9 4.39313; 790 0.5 909.9 4.11188; 791 0.75 909.9 3.83063;
792 1 909.9 3.54938; 793 1.25 909.9 3.26813; 794 1.5 909.9 2.98688;
795 1.75 909.9 2.70563; 796 -0.5 909.9 0.675; 797 -0.75 909.9 0.675;
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801 0.5 909.9 0.675; 802 0.75 909.9 0.675; 803 1 909.9 0.675;
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807 1.5 909.9 0.675; 808 -1.5 909.9 0.675; 809 -1.75 909.9 0.675;
810 -2 909.9 0.675; 811 1.75 909.9 0.675; 812 2 909.9 0.675;
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863 -2.5 909.9 0.675;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
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333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 341 319 320 316 315;
342 320 303 10 316; 343 321 322 302 303; 344 322 323 304 302;
345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306;
348 326 327 308 307; 351 330 331 320 319; 352 331 321 303 320;
383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370;
386 371 373 374 372; 387 373 375 376 374; 388 375 377 378 376;
389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386;
392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366;
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398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379;
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404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390;
407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393;
410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397;
413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400;
416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403;
419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407;
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426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415;
431 315 316 419 418; 432 316 10 409 419; 465 368 367 455 456;
466 367 370 457 455; 467 370 372 458 457; 468 372 374 459 458;
469 374 376 460 459; 470 376 378 461 460; 471 378 380 462 461;
472 380 382 463 462; 473 386 385 464 465; 474 385 368 456 464;
607 609 610 611 612; 608 610 613 614 611; 609 613 615 616 614;
615 627 628 629 630; 616 628 609 612 629; 617 631 632 610 609;
618 632 633 613 610; 619 633 634 615 613; 625 640 641 628 627;
626 641 631 609 628; 627 642 643 322 321; 628 643 644 323 322;
629 644 645 324 323; 630 645 646 325 324; 631 646 647 326 325;
635 651 652 331 330; 636 652 642 321 331; 637 612 611 643 642;
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659 666 319 315 667; 660 668 330 319 666; 661 669 383 386 670;
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736 735 722 721 734; 737 729 716 722 735; 738 732 719 723 736;
739 737 724 725 738; 740 739 726 724 737; 741 740 727 717 730;
742 738 725 727 740; 743 723 674 713 741; 744 741 713 714 742;
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748 658 747 745 656; 749 659 795 747 658; 751 657 746 750 661;
752 745 751 752 746; 753 747 753 751 745; 756 746 752 756 750;
761 661 750 761 711; 762 711 761 762 712; 763 750 756 763 761;
764 761 763 764 762; 765 701 700 765 766; 766 700 702 767 765;
767 702 703 768 767; 768 703 704 769 768; 769 704 705 770 769;
770 705 706 771 770; 771 706 707 772 771; 772 707 708 773 772;
773 710 709 774 775; 774 709 701 766 774; 775 708 712 776 773;
776 714 710 775 777; 777 742 714 777 778; 778 744 742 778 779;
779 712 762 780 776; 780 762 764 781 780; 796 615 789 618 616; 797 634 789 615;
798 789 790 618; 799 792 793 313; 800 791 792 327; 801 790 791 647;
804 794 795 659; 805 795 753 747; 806 417 794 659 406; 807 793 794 417;
808 766 765 796 797; 809 765 767 798 796; 810 767 768 799 798;
811 768 769 800 799; 812 769 770 801 800; 813 770 771 802 801;
814 771 772 803 802; 815 772 773 804 803; 816 775 774 805 806;
817 774 766 797 805; 818 773 776 807 804; 819 777 775 806 808;
820 778 777 808 809; 821 779 778 809 810; 822 776 780 811 807;
823 780 781 812 811; 840 830 728 729 831; 841 832 730 728 830;
842 833 731 732 834; 843 835 733 731 833; 844 836 734 733 835;
845 837 735 734 836; 846 831 729 735 837; 847 834 732 736 838;
848 839 737 738 840; 849 841 739 737 839; 850 842 740 730 832;
851 840 738 740 842; 852 843 830 831 844; 853 845 832 830 843;
854 846 833 834 847; 855 848 835 833 846; 856 849 836 835 848;
857 850 837 836 849; 858 844 831 837 850; 859 847 834 838 851;
860 852 839 840 853; 861 854 841 839 852; 862 855 842 832 845;
863 853 840 842 855; 864 838 736 743 856; 865 856 743 744 857;
866 851 838 856 858; 867 858 856 857 859; 868 857 744 779 860;
869 859 857 860 861; 870 860 779 810 862; 871 861 860 862 863;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 -
466 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 -
654 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 796 -
797 TO 801 804 TO 823 840 TO 871
_RAFT_MOM 326 327 333 334 337 341 TO 348 351 352 383 TO 424 427 TO 429 431 -
432 465 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 -
649 TO 651 654 659 TO 670 683 TO 685 692 695 TO 749 751 TO 753 756 -
761 TO 780 796 TO 801 804 TO 823 840 TO 871
_RAFT_SHEAR 341 351 383 TO 392 400 401 410 411 420 421 431 465 TO 474 607 -
608 TO 609 615 TO 619 625 626 635 645 649 TO 651 654 659 TO 670 692 693 695 -
696 TO 749 751 TO 753 756 761 TO 780 796 TO 799 804 TO 823 840 TO 871
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
796 TO 801 804 TO 823 840 TO 871 THICKNESS 0.75
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
796 TO 801 804 TO 823 840 TO 871 PLATE MAT DIRECT Y SUBGRADE 45600 PRINT
*******************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
```



```
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
796 TO 801 804 TO 823 840 TO 871 PR GY -30
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
796 TO 801 804 TO 823 840 TO 871 PR GY -20
JOINT LOAD
128 FX 9.664 FY -72.134 FZ 1.329
*******************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX 30.441 FY -263.577 FZ 2.183
*******************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 66.861 FY -128.014 FZ 4.007
*******************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX -66.861 FY 128.014 FZ -4.007
*******************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 6.718 FY -227.583 FZ 96.675
*******************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX -6.718 FY 227.583 FZ 96.675
*******************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL + 1.7LL
1 1.4 2 1.7
LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
1 1.4 3 1.6 2 1.0
LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
1 1.4 4 1.6 2 1.0
LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
1 1.4 5 1.6 2 1.0
LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
1 1.4 6 1.6 2 1.0
LOAD COMB 107 0.9DL + 1.6WLX1
1 0.9 3 1.6
LOAD COMB 108 0.9DL + 1.6WLX2
1 0.9 4 1.6
LOAD COMB 109 0.9DL + 1.6WLZ1
1 0.9 5 1.6
LOAD COMB 110 0.9DL + 1.6WLZ2
1 0.9 6 1.6
*******************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
```

```
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 -
431 432 465 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 -
650 TO 651 654 659 TO 670 695 TO 749 751 TO 753 756 761 TO 780 808 TO 823 -
840 TO 871
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F2_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 911.85 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.7 ZD 0.7
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 9.664 FY -72.134 FZ 1.329
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX 30.441 FY -263.577 FZ 2.183
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 66.861 FY -128.014 FZ 4.007
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX -66.861 FY 128.014 FZ -4.007
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 6.718 FY -227.583 FZ 96.675
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX -6.718 FY 227.583 FZ 96.675
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
```



```
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 4 STAAD INPUT – FOUNDATION F3

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175;
41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 911.85 3.175;
302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438;
306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438;
309 1 909.9 3.175; 311 1.275 909.9 3.175; 313 1 909.9 3.42438;
314 1.275 909.9 3.42438; 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438;
324 0 909.9 3.67438; 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438;
327 0.75 909.9 3.67438; 328 1 909.9 3.67438; 329 1.275 909.9 3.67438;
399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438;
402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438;
405 1 909.9 2.67438; 406 1.275 909.9 2.67438; 410 -0.5 909.9 2.92438;
411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438;
414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438;
417 1.275 909.9 2.92438; 420 1.525 909.9 3.42438; 421 1.525 909.9 3.175;
422 1.525 909.9 3.67438; 429 1.525 909.9 2.67438; 430 1.525 909.9 2.92438;
468 1.775 909.9 3.42438; 469 1.775 909.9 3.175; 470 1.775 909.9 3.67438;
477 1.775 909.9 2.67438; 478 1.775 909.9 2.92438; 479 1.275 911.85 3.175;
MEMBER INCIDENCES
185 2 128; 487 311 479;
ELEMENT INCIDENCES SHELL
327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41;
334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309; 344 322 323 304 302;
345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306;
348 326 327 308 307; 349 327 328 313 308; 350 328 329 314 313;
414 410 411 400 399; 415 411 412 401 400; 416 412 413 402 401;
417 413 414 403 402; 418 414 415 404 403; 419 415 416 405 404;
420 416 417 406 405; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412;
427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 430 309 311 417 416;
433 314 420 421 311; 434 329 422 420 314; 441 417 430 429 406;
442 311 421 430 417; 477 420 468 469 421; 478 422 470 468 420;
485 430 478 477 429; 486 421 469 478 430;
START GROUP DEFINITION
ELEMENT
_RAFT 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 -
441 442 477 478 485 486
_RAFT_MOM 327 333 334 337 344 TO 350 414 TO 420 424 427 TO 429 434 441 477 -
478 485 486
_RAFT_SHEAR 327 333 334 337 344 TO 350 414 TO 420 424 427 TO 429 434 441 477 -
478 485 486
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
477 478 485 486 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 487 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
477 478 485 486 PLATE MAT DIRECT Y SUBGRADE 45600 PRINT
*********************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
```



```
327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
477 478 485 486 PR GY -15
* LOAD DUE TO SURCHARGE
327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
477 478 485 486 PR GY -20
JOINT LOAD
479 FX 0.131 FY -3.735 FZ -0.041
128 FX -0.131 FY -3.507 FZ -0.121
*********************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
479 FX 0.43 FY -11.123 FZ -0.168
128 FX -0.43 FY -10.819 FZ -0.501
*********************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
479 FX 1.349 FY -2.444 FZ 0
128 FX 3.963 FY 2.444 FZ 0.003
*********************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
479 FX -1.349 FY 2.444 FZ 0
128 FX -3.963 FY -2.444 FZ -0.003
*********************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
479 FX 0.063 FY 0.218 FZ 0.603
128 FX 0.199 FY 0.455 FZ 0.786
*********************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
479 FX -0.063 FY -0.218 FZ -0.603
128 FX -0.199 FY -0.455 FZ -0.786
*********************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL + 1.7LL
1 1.4 2 1.7
LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
1 1.4 3 1.6 2 1.0
LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
1 1.4 4 1.6 2 1.0
LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
1 1.4 5 1.6 2 1.0
LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
1 1.4 6 1.6 2 1.0
LOAD COMB 107 0.9DL + 1.6WLX1
1 0.9 3 1.6
LOAD COMB 108 0.9DL + 1.6WLX2
1 0.9 4 1.6
LOAD COMB 109 0.9DL + 1.6WLZ1
1 0.9 5 1.6
LOAD COMB 110 0.9DL + 1.6WLZ2
1 0.9 6 1.6
*********************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
```

```
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 487
CLB 0.05 MEMB 185 487
CLS 0.05 ALL
CLT 0.05 MEMB 185 487
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 487
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F3-STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 909.9 3.175; 128 0 911.85 3.175; 311 1.275 909.9 3.175;
479 1.275 911.85 3.175;
MEMBER INCIDENCES
185 2 128; 487 311 479;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 487 PRIS YD 0.5 ZD 0.5
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 311 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
479 FX 0.131 FY -3.735 FZ -0.041
128 FX -0.131 FY -3.507 FZ -0.121
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
479 FX 0.43 FY -11.123 FZ -0.168
128 FX -0.43 FY -10.819 FZ -0.501
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
479 FX 1.349 FY -2.444 FZ 0
128 FX 3.963 FY 2.444 FZ 0.003
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
479 FX -1.349 FY 2.444 FZ 0
128 FX -3.963 FY -2.444 FZ -0.003
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
479 FX 0.063 FY 0.218 FZ 0.603
128 FX 0.199 FY 0.455 FZ 0.786
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
479 FX -0.063 FY -0.218 FZ -0.603
128 FX -0.199 FY -0.455 FZ -0.786
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
```



```
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 5 STAAD OUTPUT – FOUNDATION F1

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```
*****************************************************
*                                                   *
*          STAAD.Pro V8i SELECTseries5              *
*          Version  20.07.10.64                     *
*          Proprietary Program of                   *
*          Bentley Systems, Inc.                    *
*          Date=    MAY 20, 2020                    *
*          Time=    17:18:28                        *
*                                                   *
*     USER ID: LNT ECC DIV                          *
*****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1 DESIGN.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 913.6 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 311 1.25 909.9 3.175; 313 1 909.9 3.42438; 314 1.25 909.9 3.42438
   19. 315 -1.25 909.9 3.175; 316 -1 909.9 3.175; 319 -1.25 909.9 3.42438
   20. 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438; 322 -0.5 909.9 3.67438
   21. 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438; 325 0.25 909.9 3.67438
   22. 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438; 328 1 909.9 3.67438
   23. 329 1.25 909.9 3.67438; 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438
   24. 332 -0.75 909.9 3.92438; 333 -0.5 909.9 3.92438; 334 -0.25 909.9 3.92438
   25. 335 0 909.9 3.92438; 336 0.25 909.9 3.92438; 337 0.5 909.9 3.92438
   26. 338 0.75 909.9 3.92438; 339 1 909.9 3.92438; 340 1.25 909.9 3.92438
   27. 341 -1.25 909.9 3.92438; 342 -1 909.9 3.92438; 343 -0.75 909.9 4.17438
   28. 344 -0.5 909.9 4.17438; 345 -0.25 909.9 4.17438; 346 0 909.9 4.17438
   29. 347 0.25 909.9 4.17438; 348 0.5 909.9 4.17438; 349 0.75 909.9 4.17438
   30. 350 1 909.9 4.17438; 351 1.25 909.9 4.17438; 352 -1.25 909.9 4.17438
   31. 353 -1 909.9 4.17438; 354 -0.75 909.9 4.42438; 355 -0.5 909.9 4.42438
   32. 356 -0.25 909.9 4.42438; 357 0 909.9 4.42438; 358 0.25 909.9 4.42438
   33. 359 0.5 909.9 4.42438; 360 0.75 909.9 4.42438; 361 1 909.9 4.42438
   34. 362 1.25 909.9 4.42438; 363 -1.25 909.9 4.42438; 364 -1 909.9 4.42438
   35. 365 -0.75 909.9 2.17438; 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925
   36. 368 -0.75 909.9 1.925; 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925
   37. 371 0 909.9 2.17438; 372 0 909.9 1.925; 373 0.25 909.9 2.17438
   38. 374 0.25 909.9 1.925; 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925
```



```
   39. 377 0.75 909.9 2.17438; 378 0.75 909.9 1.925; 379 1 909.9 2.17438
   40. 380 1 909.9 1.925; 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925
   41. 383 -1.25 909.9 2.17438; 384 -1 909.9 2.17438; 385 -1 909.9 1.925
   42. 386 -1.25 909.9 1.925; 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438
   43. 389 -0.25 909.9 2.42438; 390 0 909.9 2.42438; 391 0.25 909.9 2.42438
   44. 392 0.5 909.9 2.42438; 393 0.75 909.9 2.42438; 394 1 909.9 2.42438
   45. 395 1.25 909.9 2.42438; 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438
   46. 398 -0.75 909.9 2.67438; 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438
   47. 401 0 909.9 2.67438; 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438
   48. 404 0.75 909.9 2.67438; 405 1 909.9 2.67438; 406 1.25 909.9 2.67438
   49. 407 -1.25 909.9 2.67438; 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438
   50. 410 -0.5 909.9 2.92438; 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438
   51. 413 0.25 909.9 2.92438; 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438
   52. 416 1 909.9 2.92438; 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438
   53. 419 -1 909.9 2.92438; 420 1.5 909.9 3.42438; 421 1.5 909.9 3.175
   54. 422 1.5 909.9 3.67438; 423 1.5 909.9 3.92438; 424 1.5 909.9 4.17438
   55. 425 1.5 909.9 4.42438; 426 1.5 909.9 2.17438; 427 1.5 909.9 1.925
   56. 428 1.5 909.9 2.42438; 429 1.5 909.9 2.67438; 430 1.5 909.9 2.92438
   57. 431 -1.5 909.9 3.42438; 432 -1.5 909.9 3.175; 433 -1.5 909.9 3.67438
   58. 434 -1.5 909.9 3.92438; 435 -1.5 909.9 4.17438; 436 -1.5 909.9 4.42438
   59. 437 -1.5 909.9 2.17438; 438 -1.5 909.9 1.925; 439 -1.5 909.9 2.42438
   60. 440 -1.5 909.9 2.67438; 441 -1.5 909.9 2.92438; 442 -0.75 909.9 4.67438
   61. 443 -0.5 909.9 4.67438; 444 -0.25 909.9 4.67438; 445 0 909.9 4.67438
   62. 446 0.25 909.9 4.67438; 447 0.5 909.9 4.67438; 448 0.75 909.9 4.67438
   63. 449 1 909.9 4.67438; 450 1.25 909.9 4.67438; 451 -1.25 909.9 4.67438
   64. 452 -1 909.9 4.67438; 453 1.5 909.9 4.67438; 454 -1.5 909.9 4.67438
   65. 455 -0.5 909.9 1.675; 456 -0.75 909.9 1.675; 457 -0.25 909.9 1.675
   66. 458 0 909.9 1.675; 459 0.25 909.9 1.675; 460 0.5 909.9 1.675
   67. 461 0.75 909.9 1.675; 462 1 909.9 1.675; 463 1.25 909.9 1.675
   68. 464 -1 909.9 1.675; 465 -1.25 909.9 1.675; 466 1.5 909.9 1.675
   69. 467 -1.5 909.9 1.675; 468 1.75 909.9 3.42438; 469 1.75 909.9 3.175
   70. 470 1.75 909.9 3.67438; 471 1.75 909.9 3.92438; 472 1.75 909.9 4.17438
   71. 473 1.75 909.9 4.42438; 474 1.75 909.9 2.17438; 475 1.75 909.9 1.925
   72. 476 1.75 909.9 2.42438; 477 1.75 909.9 2.67438; 478 1.75 909.9 2.92438
   73. 479 1.75 909.9 4.67438; 480 1.75 909.9 1.675; 481 -1.75 909.9 3.42438
   74. 482 -1.75 909.9 3.175; 483 -1.75 909.9 3.67438; 484 -1.75 909.9 3.92438
   75. 485 -1.75 909.9 4.17438; 486 -1.75 909.9 4.42438; 487 -1.75 909.9 2.17438
   76. 488 -1.75 909.9 1.925; 489 -1.75 909.9 2.42438; 490 -1.75 909.9 2.67438
   77. 491 -1.75 909.9 2.92438; 492 -1.75 909.9 4.67438; 493 -1.75 909.9 1.675
   78. 494 -0.5 909.9 1.425; 495 -0.75 909.9 1.425; 496 -0.25 909.9 1.425
   79. 497 0 909.9 1.425; 498 0.25 909.9 1.425; 499 0.5 909.9 1.425
   80. 500 0.75 909.9 1.425; 501 1 909.9 1.425; 502 1.25 909.9 1.425
   81. 503 -1 909.9 1.425; 504 -1.25 909.9 1.425; 505 1.5 909.9 1.425
   82. 506 -1.5 909.9 1.425; 507 1.75 909.9 1.425; 508 -1.75 909.9 1.425
   83. 509 -0.75 909.9 4.92438; 510 -0.5 909.9 4.92438; 511 -0.25 909.9 4.92438
   84. 512 0 909.9 4.92438; 513 0.25 909.9 4.92438; 514 0.5 909.9 4.92438
   85. 515 0.75 909.9 4.92438; 516 1 909.9 4.92438; 517 1.25 909.9 4.92438
   86. 518 -1.25 909.9 4.92438; 519 -1 909.9 4.92438; 520 1.5 909.9 4.92438
   87. 521 -1.5 909.9 4.92438; 522 1.75 909.9 4.92438; 523 -1.75 909.9 4.92438
   88. 524 2 909.9 3.42438; 525 2 909.9 3.175; 526 2 909.9 3.67438
   89. 527 2 909.9 3.92438; 528 2 909.9 4.17438; 529 2 909.9 4.42438
   90. 530 2 909.9 2.17438; 531 2 909.9 1.925; 532 2 909.9 2.42438
   91. 533 2 909.9 2.67438; 534 2 909.9 2.92438; 535 2 909.9 4.67438
   92. 536 2 909.9 1.675; 537 2 909.9 1.425; 538 2 909.9 4.92438
   93. 539 -2 909.9 3.42438; 540 -2 909.9 3.175; 541 -2 909.9 3.67438
   94. 542 -2 909.9 3.92438; 543 -2 909.9 4.17438; 544 -2 909.9 4.42438
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 95. 545 -2 909.9 2.17438; 546 -2 909.9 1.925; 547 -2 909.9 2.42438
 96. 548 -2 909.9 2.67438; 549 -2 909.9 2.92438; 550 -2 909.9 4.67438
 97. 551 -2 909.9 1.675; 552 -2 909.9 1.425; 553 -2 909.9 4.92438
 98. 554 -0.5 909.9 1.175; 555 -0.75 909.9 1.175; 556 -0.25 909.9 1.175
 99. 557 0 909.9 1.175; 558 0.25 909.9 1.175; 559 0.5 909.9 1.175
100. 560 0.75 909.9 1.175; 561 1 909.9 1.175; 562 1.25 909.9 1.175
101. 563 -1 909.9 1.175; 564 -1.25 909.9 1.175; 565 1.5 909.9 1.175
102. 566 -1.5 909.9 1.175; 567 1.75 909.9 1.175; 568 -1.75 909.9 1.175
103. 569 2 909.9 1.175; 570 -2 909.9 1.175; 571 -0.75 909.9 5.17438
104. 572 -0.5 909.9 5.17438; 573 -0.25 909.9 5.17438; 574 0 909.9 5.17438
105. 575 0.25 909.9 5.17438; 576 0.5 909.9 5.17438; 577 0.75 909.9 5.17438
106. 578 1 909.9 5.17438; 579 1.25 909.9 5.17438; 580 -1.25 909.9 5.17438
107. 581 -1 909.9 5.17438; 582 1.5 909.9 5.17438; 583 -1.5 909.9 5.17438
108. 584 1.75 909.9 5.17438; 585 -1.75 909.9 5.17438; 586 2 909.9 5.17438
109. 587 -2 909.9 5.17438
110. MEMBER INCIDENCES
111. 185 2 128
112. ELEMENT INCIDENCES SHELL
113. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
114. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309
115. 341 319 320 316 315; 342 320 303 10 316; 343 321 322 302 303
116. 344 322 323 304 302; 345 323 324 305 304; 346 324 325 306 305
117. 347 325 326 307 306; 348 326 327 308 307; 349 327 328 313 308
118. 350 328 329 314 313; 351 330 331 320 319; 352 331 321 303 320
119. 353 332 333 322 321; 354 333 334 323 322; 355 334 335 324 323
120. 356 335 336 325 324; 357 336 337 326 325; 358 337 338 327 326
121. 359 338 339 328 327; 360 339 340 329 328; 361 341 342 331 330
122. 362 342 332 321 331; 363 343 344 333 332; 364 344 345 334 333
123. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
124. 368 348 349 338 337; 369 349 350 339 338; 370 350 351 340 339
125. 371 352 353 342 341; 372 353 343 332 342; 373 354 355 344 343
126. 374 355 356 345 344; 375 356 357 346 345; 376 357 358 347 346
127. 377 358 359 348 347; 378 359 360 349 348; 379 360 361 350 349
128. 380 361 362 351 350; 381 363 364 353 352; 382 364 354 343 353
129. 383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370
130. 386 371 373 374 372; 387 373 375 376 374; 388 375 377 378 376
131. 389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386
132. 392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366
133. 395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373
134. 398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379
135. 401 396 397 384 383; 402 397 387 365 384; 403 398 399 388 387
136. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
137. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
138. 410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397
139. 413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400
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141. 419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407
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144. 430 309 311 417 416; 431 315 316 419 418; 432 316 10 409 419
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146. 436 351 424 423 340; 437 362 425 424 351; 438 381 426 427 382
147. 439 395 428 426 381; 440 406 429 428 395; 441 417 430 429 406
148. 442 311 421 430 417; 443 431 319 315 432; 444 433 330 319 431
149. 445 434 341 330 433; 446 435 352 341 434; 447 436 363 352 435
150. 448 437 383 386 438; 449 439 396 383 437; 450 440 407 396 439
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151. 451 441 418 407 440; 452 432 315 418 441; 453 442 443 355 354
152. 454 443 444 356 355; 455 444 445 357 356; 456 445 446 358 357
153. 457 446 447 359 358; 458 447 448 360 359; 459 448 449 361 360
154. 460 449 450 362 361; 461 451 452 364 363; 462 452 442 354 364
155. 463 450 453 425 362; 464 454 451 363 436; 465 368 367 455 456
156. 466 367 370 457 455; 467 370 372 458 457; 468 372 374 459 458
157. 469 374 376 460 459; 470 376 378 461 460; 471 378 380 462 461
158. 472 380 382 463 462; 473 386 385 464 465; 474 385 368 456 464
159. 475 382 427 466 463; 476 438 386 465 467; 477 420 468 469 421
160. 478 422 470 468 420; 479 423 471 470 422; 480 424 472 471 423
161. 481 425 473 472 424; 482 426 474 475 427; 483 428 476 474 426
162. 484 429 477 476 428; 485 430 478 477 429; 486 421 469 478 430
163. 487 453 479 473 425; 488 427 475 480 466; 489 481 431 432 482
164. 490 483 433 431 481; 491 484 434 433 483; 492 485 435 434 484
165. 493 486 436 435 485; 494 487 437 438 488; 495 489 439 437 487
166. 496 490 440 439 489; 497 491 441 440 490; 498 482 432 441 491
167. 499 492 454 436 486; 500 488 438 467 493; 501 456 455 494 495
168. 502 455 457 496 494; 503 457 458 497 496; 504 458 459 498 497
169. 505 459 460 499 498; 506 460 461 500 499; 507 461 462 501 500
170. 508 462 463 502 501; 509 465 464 503 504; 510 464 456 495 503
171. 511 463 466 505 502; 512 467 465 504 506; 513 466 480 507 505
172. 514 493 467 506 508; 515 509 510 443 442; 516 510 511 444 443
173. 517 511 512 445 444; 518 512 513 446 445; 519 513 514 447 446
174. 520 514 515 448 447; 521 515 516 449 448; 522 516 517 450 449
175. 523 518 519 452 451; 524 519 509 442 452; 525 517 520 453 450
176. 526 521 518 451 454; 527 520 522 479 453; 528 523 521 454 492
177. 529 468 524 525 469; 530 470 526 524 468; 531 471 527 526 470
178. 532 472 528 527 471; 533 473 529 528 472; 534 474 530 531 475
179. 535 476 532 530 474; 536 477 533 532 476; 537 478 534 533 477
180. 538 469 525 534 478; 539 479 535 529 473; 540 475 531 536 480
181. 541 480 536 537 507; 542 522 538 535 479; 543 539 481 482 540
182. 544 541 483 481 539; 545 542 484 483 541; 546 543 485 484 542
183. 547 544 486 485 543; 548 545 487 488 546; 549 547 489 487 545
184. 550 548 490 489 547; 551 549 491 490 548; 552 540 482 491 549
185. 553 550 492 486 544; 554 546 488 493 551; 555 551 493 508 552
186. 556 553 523 492 550; 557 495 494 554 555; 558 494 496 556 554
187. 559 496 497 557 556; 560 497 498 558 557; 561 498 499 559 558
188. 562 499 500 560 559; 563 500 501 561 560; 564 501 502 562 561
189. 565 504 503 563 564; 566 503 495 555 563; 567 502 505 565 562
190. 568 506 504 564 566; 569 505 507 567 565; 570 508 506 566 568
191. 571 507 537 569 567; 572 552 508 568 570; 573 571 572 510 509
192. 574 572 573 511 510; 575 573 574 512 511; 576 574 575 513 512
193. 577 575 576 514 513; 578 576 577 515 514; 579 577 578 516 515
194. 580 578 579 517 516; 581 580 581 519 518; 582 581 571 509 519
195. 583 579 582 520 517; 584 583 580 518 521; 585 582 584 522 520
196. 586 585 583 521 523; 587 584 586 538 522; 588 587 585 523 553
197. START GROUP DEFINITION
198. ELEMENT
199. _RAFT 326 TO 328 332 TO 334 337 338 341 TO 588
200. _RAFT_MOM 326 327 333 334 337 338 341 TO 424 427 TO 588
201. _RAFT_SHEAR 338 341 350 351 360 361 370 371 373 TO 392 400 401 410 411 420 -
202. 421 430 431 433 TO 588
203. MEMBER
204. END GROUP DEFINITION
205. ELEMENT PROPERTY
206. 326 TO 328 332 TO 334 337 338 341 TO 588 THICKNESS 0.75
```

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207. DEFINE MATERIAL START
208. ISOTROPIC CONCRETE
209. E 2.17185E+007
210. POISSON 0.17
211. DENSITY 25
212. ALPHA 1E-005
213. DAMP 0.05
214. TYPE CONCRETE
215. STRENGTH FCU 27579
216. END DEFINE MATERIAL
217. MEMBER PROPERTY AMERICAN
218. 185 PRIS YD 0.7 ZD 0.7
219. CONSTANTS
220. MATERIAL CONCRETE ALL
221. SUPPORTS
222. 326 TO 328 332 TO 334 337 338 341 TO 587 -
223. 588 PLATE MAT DIRECT Y SUBGRADE 57000 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
10	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
303	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.062500
311	0.062500
313	0.062423
314	0.062422
315	0.062500
316	0.062500
319	0.062422
320	0.062423
321	0.062500
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406	0.062500
407	0.062500
408	0.062500
409	0.062578
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.062578
417	0.062578
418	0.062578
419	0.062578
420	0.062422
421	0.062500
422	0.062500
423	0.062500
424	0.062500
425	0.062500
426	0.062422
427	0.062422
428	0.062500
429	0.062500
430	0.062577
431	0.062422
432	0.062500
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437	0.062422
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579	0.031250
580	0.031250
581	0.031250
582	0.031250
583	0.031250
584	0.031250
585	0.031250
586	0.015625
587	0.015625
TOTAL	15.997518

```
224. ********************************************************************************
225. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
226. SELFWEIGHT Y -1
227. ELEMENT LOAD
228. * LOAD DUE TO BACKFILL
229. 326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -47
230. * LOAD DUE TO SURCHARGE
231. 326 TO 328 332 TO 334 337 338 341 TO 588 PR GY -20
232. JOINT LOAD
233. 128 FX -25.516 FY -145.216 FZ 0.208
234. ********************************************************************************
235. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
236. JOINT LOAD
237. 128 FX -58.516 FY -379.137 FZ -1.923
238. ********************************************************************************
239. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
240. JOINT LOAD
241. 128 FX 89.549 FY 82.052 FZ -4.011
242. ********************************************************************************
243. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
244. JOINT LOAD
245. 128 FX -89.549 FY -82.052 FZ 4.011
246. ********************************************************************************
247. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
248. JOINT LOAD
```

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249. 128 FX -16.504 FY -361.777 FZ 103.166
250. **************************************************************************
251. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
252. JOINT LOAD
253. 128 FX 16.504 FY 361.777 FZ -103.166
254. **************************************************************************
255. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
256. LOAD COMB 101 1.4DL
257. 1 1.4
258. LOAD COMB 102 1.4DL + 1.7LL
259. 1 1.4 2 1.7
260. LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
261. 1 1.4 3 1.6 2 1.0
262. LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
263. 1 1.4 4 1.6 2 1.0
264. LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
265. 1 1.4 5 1.6 2 1.0
266. LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
267. 1 1.4 6 1.6 2 1.0
268. LOAD COMB 107 0.9DL + 1.6WLX1
269. 1 0.9 3 1.6
270. LOAD COMB 108 0.9DL + 1.6WLX2
271. 1 0.9 4 1.6
272. LOAD COMB 109 0.9DL + 1.6WLZ1
273. 1 0.9 5 1.6
274. LOAD COMB 110 0.9DL + 1.6WLZ2
275. 1 0.9 6 1.6
276. **************************************************************************
277. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
278. LOAD COMB 201 1.0DL
279. 1 1.0
280. LOAD COMB 202 1.0DL + 1.0LL
281. 1 1.0 2 1.0
282. LOAD COMB 203 1.0DL + 1.0WLX1
283. 1 1.0 3 1.0
284. LOAD COMB 204 1.0DL + 1.0WLX2
285. 1 1.0 4 1.0
286. LOAD COMB 205 1.0DL + 1.0WLZ1
287. 1 1.0 5 1.0
288. LOAD COMB 206 1.0DL + 1.0WLZ2
289. 1 1.0 6 1.0
290. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
291. 1 1.0 3 0.75 2 0.75
292. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
293. 1 1.0 4 0.75 2 0.75
294. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
295. 1 1.0 5 0.75 2 0.75
296. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
297. 1 1.0 6 0.75 2 0.75
298. LOAD COMB 211 0.6DL + 1.0WLX1
299. 1 0.6 3 1.0
300. LOAD COMB 212 0.6DL + 1.0WLX2
301. 1 0.6 4 1.0
302. LOAD COMB 213 0.6DL + 1.0WLZ1
303. 1 0.6 5 1.0
304. LOAD COMB 214 0.6DL + 1.0WLZ2
```



```
305. 1 0.6 6 1.0
306. **************************************************************************
307. PERFORM ANALYSIS


         P R O B L E M   S T A T I S T I C S
         -----------------------------------

  NUMBER OF JOINTS        290  NUMBER OF MEMBERS       1
  NUMBER OF PLATES        256  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    289

       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=   124/    29/     90 DOF
TOTAL      PRIMARY LOAD CASES =     6, TOTAL DEGREES OF FREEDOM =     873
TOTAL LOAD COMBINATION  CASES =    24  SO FAR.
  SIZE OF STIFFNESS MATRIX =       79 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     14.2/ 146365.4 MB



308. LOAD LIST 101 TO 110
309. START CONCRETE DESIGN
```

```
310. CODE ACI
311. TRACK 2 ALL
312. FC 35000 MEMB 185
313. CLB 0.05 MEMB 185
314. CLS 0.05 ALL
315. CLT 0.05 ALL
316. MAXMAIN 25 ALL
317. MINMAIN 25 ALL
318. MINSEC 10 ALL
319. DESIGN COLUMN 185
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=================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 700.0 X 700.0 MMS, TIED
       AREA OF STEEL REQUIRED =  9770.6  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

 20 - 25 MM              2.004      106      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
18346.75  14677.40   6280.15   2021.79     321.9
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1183.91  -4061.32    103.15    386.12    1.01165
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
18346.75  14677.40   6280.15   2021.79     321.9
   M0       P-tens.   Des.Pn    Des.Mn     e/h
 1183.91  -4061.32    103.15    948.88    2.48614
 ---------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |             13548.37  1240.91  6774.18  1993.06
       P0 |*            12419.34  1434.34  5645.15  2002.02
          | *           11290.31  1600.82  4516.12  1944.54
   Pn,max|__*          10161.28  1731.78  3387.09  1830.84
          |   *          9032.24  1834.95  2258.06  1668.77
    Pn    |    *         7903.21  1920.77  1129.03  1450.42
   NOMINAL|     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL |      *       13548.37  1240.91  6774.18  1993.06
COMPRESSION|       *      12419.34  1434.34  5645.15  2002.02
        Pb|-------*Mb    11290.31  1600.82  4516.12  1944.54
          |      *       10161.28  1731.78  3387.09  1830.84
__________|____*_______  9032.24  1834.95  2258.06  1668.77
          |  * M0   Mn,  7903.21  1920.77  1129.03  1450.42
          | *   BENDING
   P-tens |*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************


320. END CONCRETE DESIGN
```

321. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 20,2020 TIME= 17:18:36 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    MAY 22, 2020                *
            *           Time=    12:29:14                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1 _STABILITY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F1_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 913.6 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 PRIS YD 0.7 ZD 0.7
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FX -25.516 FY -145.216 FZ 0.208
   37. ******************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
```



```
   39. JOINT LOAD
   40. 128 FX -58.516 FY -379.137 FZ -1.923
   41. ******************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
   43. JOINT LOAD
   44. 128 FX 89.549 FY 82.052 FZ -4.011
   45. ******************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
   47. JOINT LOAD
   48. 128 FX -89.549 FY -82.052 FZ 4.011
   49. ******************************************************************************
   50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
   51. JOINT LOAD
   52. 128 FX -16.504 FY -361.777 FZ 103.166
   53. ******************************************************************************
   54. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
   55. JOINT LOAD
   56. 128 FX 16.504 FY 361.777 FZ -103.166
   57. ******************************************************************************
   58. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   59. LOAD COMB 201 1.0DL
   60. 1 1.0
   61. LOAD COMB 202 1.0DL + 1.0LL
   62. 1 1.0 2 1.0
   63. LOAD COMB 203 1.0DL + 1.0WLX1
   64. 1 1.0 3 1.0
   65. LOAD COMB 204 1.0DL + 1.0WLX2
   66. 1 1.0 4 1.0
   67. LOAD COMB 205 1.0DL + 1.0WLZ1
   68. 1 1.0 5 1.0
   69. LOAD COMB 206 1.0DL + 1.0WLZ2
   70. 1 1.0 6 1.0
   71. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
   72. 1 1.0 3 0.75 2 0.75
   73. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
   74. 1 1.0 4 0.75 2 0.75
   75. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
   76. 1 1.0 5 0.75 2 0.75
   77. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
   78. 1 1.0 6 0.75 2 0.75
   79. LOAD COMB 211 0.6DL + 1.0WLX1
   80. 1 0.6 3 1.0
   81. LOAD COMB 212 0.6DL + 1.0WLX2
   82. 1 0.6 4 1.0
   83. LOAD COMB 213 0.6DL + 1.0WLZ1
   84. 1 0.6 5 1.0
   85. LOAD COMB 214 0.6DL + 1.0WLZ2
   86. 1 0.6 6 1.0
   87. ******************************************************************************
   88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 146353.4 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 12:29:19 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 6 STAAD OUTPUT – FOUNDATION F2

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    MAY 22, 2020                 *
*          Time=    12:32:53                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2 DESIGN..STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F2
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 10 -0.75 909.9 3.175; 34 -0.5 909.9 3.175
   14. 35 -0.25 909.9 3.175; 41 0.25 909.9 3.175; 42 0.5 909.9 3.175
   15. 128 0 911.85 3.175; 302 -0.5 909.9 3.42438; 303 -0.75 909.9 3.42438
   16. 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438; 306 0.25 909.9 3.42438
   17. 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438; 309 1 909.9 3.175
   18. 313 1 909.9 3.42438; 315 -1.25 909.9 3.175; 316 -1 909.9 3.175
   19. 319 -1.25 909.9 3.42438; 320 -1 909.9 3.42438; 321 -0.75 909.9 3.67438
   20. 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438; 324 0 909.9 3.67438
   21. 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438; 327 0.75 909.9 3.67438
   22. 330 -1.25 909.9 3.67438; 331 -1 909.9 3.67438; 365 -0.75 909.9 2.17438
   23. 366 -0.5 909.9 2.17438; 367 -0.5 909.9 1.925; 368 -0.75 909.9 1.925
   24. 369 -0.25 909.9 2.17438; 370 -0.25 909.9 1.925; 371 0 909.9 2.17438
   25. 372 0 909.9 1.925; 373 0.25 909.9 2.17438; 374 0.25 909.9 1.925
   26. 375 0.5 909.9 2.17438; 376 0.5 909.9 1.925; 377 0.75 909.9 2.17438
   27. 378 0.75 909.9 1.925; 379 1 909.9 2.17438; 380 1 909.9 1.925
   28. 381 1.25 909.9 2.17438; 382 1.25 909.9 1.925; 383 -1.25 909.9 2.17438
   29. 384 -1 909.9 2.17438; 385 -1 909.9 1.925; 386 -1.25 909.9 1.925
   30. 387 -0.75 909.9 2.42438; 388 -0.5 909.9 2.42438; 389 -0.25 909.9 2.42438
   31. 390 0 909.9 2.42438; 391 0.25 909.9 2.42438; 392 0.5 909.9 2.42438
   32. 393 0.75 909.9 2.42438; 394 1 909.9 2.42438; 395 1.25 909.9 2.42438
   33. 396 -1.25 909.9 2.42438; 397 -1 909.9 2.42438; 398 -0.75 909.9 2.67438
   34. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   35. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   36. 405 1 909.9 2.67438; 406 1.25 909.9 2.67438; 407 -1.25 909.9 2.67438
   37. 408 -1 909.9 2.67438; 409 -0.75 909.9 2.92438; 410 -0.5 909.9 2.92438
   38. 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438
```



```
   39. 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438
   40. 417 1.25 909.9 2.92438; 418 -1.25 909.9 2.92438; 419 -1 909.9 2.92438
   41. 455 -0.5 909.9 1.675; 456 -0.75 909.9 1.675; 457 -0.25 909.9 1.675
   42. 458 0 909.9 1.675; 459 0.25 909.9 1.675; 460 0.5 909.9 1.675
   43. 461 0.75 909.9 1.675; 462 1 909.9 1.675; 463 1.25 909.9 1.675
   44. 464 -1 909.9 1.675; 465 -1.25 909.9 1.675; 609 -0.75 909.9 4.42438
   45. 610 -0.5 909.9 4.42438; 611 -0.5 909.9 4.175; 612 -0.75 909.9 4.175
   46. 613 -0.25 909.9 4.42438; 614 -0.25 909.9 4.175; 615 0 909.9 4.42438
   47. 616 0 909.9 4.175; 618 0.25 909.9 4.175; 627 -1.25 909.9 4.42438
   48. 628 -1 909.9 4.42438; 629 -1 909.9 4.175; 630 -1.25 909.9 4.175
   49. 631 -0.75 909.9 4.67438; 632 -0.5 909.9 4.67438; 633 -0.25 909.9 4.67438
   50. 634 0 909.9 4.67438; 640 -1.25 909.9 4.67438; 641 -1 909.9 4.67438
   51. 642 -0.75 909.9 3.92438; 643 -0.5 909.9 3.92438; 644 -0.25 909.9 3.92438
   52. 645 0 909.9 3.92438; 646 0.25 909.9 3.92438; 647 0.5 909.9 3.92438
   53. 651 -1.25 909.9 3.92438; 652 -1 909.9 3.92438; 656 1.5 909.9 2.17438
   54. 657 1.5 909.9 1.925; 658 1.5 909.9 2.42438; 659 1.5 909.9 2.67438
   55. 661 1.5 909.9 1.675; 666 -1.5 909.9 3.42438; 667 -1.5 909.9 3.175
   56. 668 -1.5 909.9 3.67438; 669 -1.5 909.9 2.17438; 670 -1.5 909.9 1.925
   57. 671 -1.5 909.9 2.42438; 672 -1.5 909.9 2.67438; 673 -1.5 909.9 2.92438
   58. 674 -1.5 909.9 1.675; 675 -1.5 909.9 4.42438; 676 -1.5 909.9 4.175
   59. 677 -1.5 909.9 4.67438; 678 -1.5 909.9 3.92438; 689 -0.5 909.9 1.425
   60. 690 -0.75 909.9 1.425; 691 -0.25 909.9 1.425; 692 0 909.9 1.425
   61. 693 0.25 909.9 1.425; 694 0.5 909.9 1.425; 695 0.75 909.9 1.425
   62. 696 1 909.9 1.425; 697 1.25 909.9 1.425; 698 -1 909.9 1.425
   63. 699 -1.25 909.9 1.425; 700 -0.5 909.9 1.175; 701 -0.75 909.9 1.175
   64. 702 -0.25 909.9 1.175; 703 0 909.9 1.175; 704 0.25 909.9 1.175
   65. 705 0.5 909.9 1.175; 706 0.75 909.9 1.175; 707 1 909.9 1.175
   66. 708 1.25 909.9 1.175; 709 -1 909.9 1.175; 710 -1.25 909.9 1.175
   67. 711 1.5 909.9 1.425; 712 1.5 909.9 1.175; 713 -1.5 909.9 1.425
   68. 714 -1.5 909.9 1.175; 715 -1.75 909.9 3.42438; 716 -1.75 909.9 3.175
   69. 717 -1.75 909.9 3.67438; 718 -1.75 909.9 2.17438; 719 -1.75 909.9 1.925
   70. 720 -1.75 909.9 2.42438; 721 -1.75 909.9 2.67438; 722 -1.75 909.9 2.92438
   71. 723 -1.75 909.9 1.675; 724 -1.75 909.9 4.42438; 725 -1.75 909.9 4.175
   72. 726 -1.75 909.9 4.67438; 727 -1.75 909.9 3.92438; 728 -2 909.9 3.42438
   73. 729 -2 909.9 3.175; 730 -2 909.9 3.67438; 731 -2 909.9 2.17438
   74. 732 -2 909.9 1.925; 733 -2 909.9 2.42438; 734 -2 909.9 2.67438
   75. 735 -2 909.9 2.92438; 736 -2 909.9 1.675; 737 -2 909.9 4.42438
   76. 738 -2 909.9 4.175; 739 -2 909.9 4.67438; 740 -2 909.9 3.92438
   77. 741 -1.75 909.9 1.425; 742 -1.75 909.9 1.175; 743 -2 909.9 1.425
   78. 744 -2 909.9 1.175; 745 1.75 909.9 2.17438; 746 1.75 909.9 1.925
   79. 747 1.75 909.9 2.42438; 750 1.75 909.9 1.675; 751 2 909.9 2.17438
   80. 752 2 909.9 1.925; 753 2 909.9 2.42438; 756 2 909.9 1.675
   81. 761 1.75 909.9 1.425; 762 1.75 909.9 1.175; 763 2 909.9 1.425
   82. 764 2 909.9 1.175; 765 -0.5 909.9 0.925; 766 -0.75 909.9 0.925
   83. 767 -0.25 909.9 0.925; 768 0 909.9 0.925; 769 0.25 909.9 0.925
   84. 770 0.5 909.9 0.925; 771 0.75 909.9 0.925; 772 1 909.9 0.925
   85. 773 1.25 909.9 0.925; 774 -1 909.9 0.925; 775 -1.25 909.9 0.925
   86. 776 1.5 909.9 0.925; 777 -1.5 909.9 0.925; 778 -1.75 909.9 0.925
   87. 779 -2 909.9 0.925; 780 1.75 909.9 0.925; 781 2 909.9 0.925
   88. 789 0.25 909.9 4.39313; 790 0.5 909.9 4.11188; 791 0.75 909.9 3.83063
   89. 792 1 909.9 3.54938; 793 1.25 909.9 3.26813; 794 1.5 909.9 2.98688
   90. 795 1.75 909.9 2.70563; 796 -0.5 909.9 0.675; 797 -0.75 909.9 0.675
   91. 798 -0.25 909.9 0.675; 799 0 909.9 0.675; 800 0.25 909.9 0.675
   92. 801 0.5 909.9 0.675; 802 0.75 909.9 0.675; 803 1 909.9 0.675
   93. 804 1.25 909.9 0.675; 805 -1 909.9 0.675; 806 -1.25 909.9 0.675
   94. 807 1.5 909.9 0.675; 808 -1.5 909.9 0.675; 809 -1.75 909.9 0.675
```

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 95. 810 -2 909.9 0.675; 811 1.75 909.9 0.675; 812 2 909.9 0.675
 96. 830 -2.25 909.9 3.42438; 831 -2.25 909.9 3.175; 832 -2.25 909.9 3.67438
 97. 833 -2.25 909.9 2.17438; 834 -2.25 909.9 1.925; 835 -2.25 909.9 2.42438
 98. 836 -2.25 909.9 2.67438; 837 -2.25 909.9 2.92438; 838 -2.25 909.9 1.675
 99. 839 -2.25 909.9 4.42438; 840 -2.25 909.9 4.175; 841 -2.25 909.9 4.67438
100. 842 -2.25 909.9 3.92438; 843 -2.5 909.9 3.42438; 844 -2.5 909.9 3.175
101. 845 -2.5 909.9 3.67438; 846 -2.5 909.9 2.17438; 847 -2.5 909.9 1.925
102. 848 -2.5 909.9 2.42438; 849 -2.5 909.9 2.67438; 850 -2.5 909.9 2.92438
103. 851 -2.5 909.9 1.675; 852 -2.5 909.9 4.42438; 853 -2.5 909.9 4.175
104. 854 -2.5 909.9 4.67438; 855 -2.5 909.9 3.92438; 856 -2.25 909.9 1.425
105. 857 -2.25 909.9 1.175; 858 -2.5 909.9 1.425; 859 -2.5 909.9 1.175
106. 860 -2.25 909.9 0.925; 861 -2.5 909.9 0.925; 862 -2.25 909.9 0.675
107. 863 -2.5 909.9 0.675
108. MEMBER INCIDENCES
109. 185 2 128
110. ELEMENT INCIDENCES SHELL
111. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
112. 333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 341 319 320 316 315
113. 342 320 303 10 316; 343 321 322 302 303; 344 322 323 304 302
114. 345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306
115. 348 326 327 308 307; 351 330 331 320 319; 352 331 321 303 320
116. 383 365 366 367 368; 384 366 369 370 367; 385 369 371 372 370
117. 386 371 373 374 372; 387 373 375 376 374; 388 375 377 378 376
118. 389 377 379 380 378; 390 379 381 382 380; 391 383 384 385 386
119. 392 384 365 368 385; 393 387 388 366 365; 394 388 389 369 366
120. 395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373
121. 398 392 393 377 375; 399 393 394 379 377; 400 394 395 381 379
122. 401 396 397 384 383; 402 397 387 365 384; 403 398 399 388 387
123. 404 399 400 389 388; 405 400 401 390 389; 406 401 402 391 390
124. 407 402 403 392 391; 408 403 404 393 392; 409 404 405 394 393
125. 410 405 406 395 394; 411 407 408 397 396; 412 408 398 387 397
126. 413 409 410 399 398; 414 410 411 400 399; 415 411 412 401 400
127. 416 412 413 402 401; 417 413 414 403 402; 418 414 415 404 403
128. 419 415 416 405 404; 420 416 417 406 405; 421 418 419 408 407
129. 422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411
130. 426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415
131. 431 315 316 419 418; 432 316 10 409 419; 465 368 367 455 456
132. 466 367 370 457 455; 467 370 372 458 457; 468 372 374 459 458
133. 469 374 376 460 459; 470 376 378 461 460; 471 378 380 462 461
134. 472 380 382 463 462; 473 386 385 464 465; 474 385 368 456 464
135. 607 609 610 611 612; 608 610 613 614 611; 609 613 615 616 614
136. 615 627 628 629 630; 616 628 609 612 629; 617 631 632 610 609
137. 618 632 633 613 610; 619 633 634 615 613; 625 640 641 628 627
138. 626 641 631 609 628; 627 642 643 322 321; 628 643 644 323 322
139. 629 644 645 324 323; 630 645 646 325 324; 631 646 647 326 325
140. 635 651 652 331 330; 636 652 642 321 331; 637 612 611 643 642
141. 638 611 614 644 643; 639 614 616 645 644; 640 616 618 646 645
142. 645 630 629 652 651; 646 629 612 642 652; 649 381 656 657 382
143. 650 395 658 656 381; 651 406 659 658 395; 654 382 657 661 463
144. 659 666 319 315 667; 660 668 330 319 666; 661 669 383 386 670
145. 662 671 396 383 669; 663 672 407 396 671; 664 673 418 407 672
146. 665 667 315 418 673; 666 670 386 465 674; 667 675 627 630 676
147. 668 677 640 627 675; 669 678 651 330 668; 670 676 630 651 678
148. 683 618 790 647 646; 684 647 791 327 326; 685 327 792 313 308; 692 313 793 309
149. 693 309 793 417 416; 695 456 455 689 690; 696 455 457 691 689
150. 697 457 458 692 691; 698 458 459 693 692; 699 459 460 694 693
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151. 700 460 461 695 694; 701 461 462 696 695; 702 462 463 697 696
152. 703 465 464 698 699; 704 464 456 690 698; 705 690 689 700 701
153. 706 689 691 702 700; 707 691 692 703 702; 708 692 693 704 703
154. 709 693 694 705 704; 710 694 695 706 705; 711 695 696 707 706
155. 712 696 697 708 707; 713 699 698 709 710; 714 698 690 701 709
156. 715 463 661 711 697; 716 697 711 712 708; 717 674 465 699 713
157. 718 713 699 710 714; 719 715 666 667 716; 720 717 668 666 715
158. 721 718 669 670 719; 722 720 671 669 718; 723 721 672 671 720
159. 724 722 673 672 721; 725 716 667 673 722; 726 719 670 674 723
160. 727 724 675 676 725; 728 726 677 675 724; 729 727 678 668 717
161. 730 725 676 678 727; 731 728 715 716 729; 732 730 717 715 728
162. 733 731 718 719 732; 734 733 720 718 731; 735 734 721 720 733
163. 736 735 722 721 734; 737 729 716 722 735; 738 732 719 723 736
164. 739 737 724 725 738; 740 739 726 724 737; 741 740 727 717 730
165. 742 738 725 727 740; 743 723 674 713 741; 744 741 713 714 742
166. 745 736 723 741 743; 746 743 741 742 744; 747 656 745 746 657
167. 748 658 747 745 656; 749 659 795 747 658; 751 657 746 750 661
168. 752 745 751 752 746; 753 747 753 751 745; 756 746 752 756 750
169. 761 661 750 761 711; 762 711 761 762 712; 763 750 756 763 761
170. 764 761 763 764 762; 765 701 700 765 766; 766 700 702 767 765
171. 767 702 703 768 767; 768 703 704 769 768; 769 704 705 770 769
172. 770 705 706 771 770; 771 706 707 772 771; 772 707 708 773 772
173. 773 710 709 774 775; 774 709 701 766 774; 775 708 712 776 773
174. 776 714 710 775 777; 777 742 714 777 778; 778 744 742 778 779
175. 779 712 762 780 776; 780 762 764 781 780; 796 615 789 618 616; 797 634 789 615
176. 798 789 790 618; 799 792 793 313; 800 791 792 327; 801 790 791 647
177. 804 794 795 659; 805 795 753 747; 806 417 794 659 406; 807 793 794 417
178. 808 766 765 796 797; 809 765 767 798 796; 810 767 768 799 798
179. 811 768 769 800 799; 812 769 770 801 800; 813 770 771 802 801
180. 814 771 772 803 802; 815 772 773 804 803; 816 775 774 805 806
181. 817 774 766 797 805; 818 773 776 807 804; 819 777 775 806 808
182. 820 778 777 808 809; 821 779 778 809 810; 822 776 780 811 807
183. 823 780 781 812 811; 840 830 728 729 831; 841 832 730 728 830
184. 842 833 731 732 834; 843 835 733 731 833; 844 836 734 733 835
185. 845 837 735 734 836; 846 831 729 735 837; 847 834 732 736 838
186. 848 839 737 738 840; 849 841 739 737 839; 850 842 740 730 832
187. 851 840 738 740 842; 852 843 830 831 844; 853 845 832 830 843
188. 854 846 833 834 847; 855 848 835 833 846; 856 849 836 835 848
189. 857 850 837 836 849; 858 844 831 837 850; 859 847 834 838 851
190. 860 852 839 840 853; 861 854 841 839 852; 862 855 842 832 845
191. 863 853 840 842 855; 864 838 736 743 856; 865 856 743 744 857
192. 866 851 838 856 858; 867 858 856 857 859; 868 857 744 779 860
193. 869 859 857 860 861; 870 860 779 810 862; 871 861 860 862 863
194. START GROUP DEFINITION
195. ELEMENT
196. _RAFT 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 -
197. 466 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 -
198. 654 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 796 -
199. 797 TO 801 804 TO 823 840 TO 871
200. _RAFT_MOM 326 327 333 334 337 341 TO 348 351 352 383 TO 424 427 TO 429 431 -
201. 432 465 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 -
202. 649 TO 651 654 659 TO 670 683 TO 685 692 695 TO 749 751 TO 753 756 -
203. 761 TO 780 796 TO 801 804 TO 823 840 TO 871
204. _RAFT_SHEAR 341 351 383 TO 392 400 401 410 411 420 421 431 465 TO 474 607 -
205. 608 TO 609 615 TO 619 625 626 635 645 649 TO 651 654 659 TO 670 692 693 695 -
206. 696 TO 749 751 TO 753 756 761 TO 780 796 TO 799 804 TO 823 840 TO 871
```

```
207. MEMBER
208. END GROUP DEFINITION
209. ELEMENT PROPERTY
210. 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
211. 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
212. 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
213. 796 TO 801 804 TO 823 840 TO 871 THICKNESS 0.75
214. DEFINE MATERIAL START
215. ISOTROPIC CONCRETE
216. E 2.17185E+007
217. POISSON 0.17
218. DENSITY 25
219. ALPHA 1E-005
220. DAMP 0.05
221. TYPE CONCRETE
222. STRENGTH FCU 27579
223. END DEFINE MATERIAL
224. MEMBER PROPERTY AMERICAN
225. 185 PRIS YD 0.7 ZD 0.7
226. CONSTANTS
227. MATERIAL CONCRETE ALL
228. SUPPORTS
229. 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
230. 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
231. 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
232. 796 TO 801 804 TO 823 840 TO 871 PLATE MAT DIRECT Y SUBGRADE 45600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
10	0.062500
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
303	0.062422
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.060469
309	0.058760
313	0.040951
315	0.062500
316	0.062500
319	0.062422
320	0.062423
321	0.062500
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.061035

327	0.047038
330	0.062500
331	0.062500
365	0.062422
366	0.062422
367	0.062422
368	0.062422
369	0.062422
370	0.062422
371	0.062422
372	0.062422
373	0.062422
374	0.062422
375	0.062422
376	0.062422
377	0.062422
378	0.062422
379	0.062423
380	0.062423
381	0.062422
382	0.062422
383	0.062422
384	0.062423
385	0.062423
386	0.062422
387	0.062500
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.062500
394	0.062500
395	0.062500
396	0.062500
397	0.062500
398	0.062500
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.062500
405	0.062500
406	0.063477
407	0.062500
408	0.062500
409	0.062578
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.064033
417	0.066579

418	0.062578
419	0.062578
455	0.062500
456	0.062500
457	0.062500
458	0.062500
459	0.062500
460	0.062500
461	0.062500
462	0.062500
463	0.062500
464	0.062500
465	0.062500
609	0.062422
610	0.062422
611	0.062500
612	0.062500
613	0.062422
614	0.062500
615	0.056726
616	0.062012
618	0.053551
627	0.062422
628	0.062423
629	0.062500
630	0.062500
631	0.031250
632	0.031250
633	0.031250
634	0.026042
640	0.031250
641	0.031250
642	0.062578
643	0.062578
644	0.062578
645	0.062578
646	0.061591
647	0.050303
651	0.062578
652	0.062578
656	0.062422
657	0.062422
658	0.062988
659	0.063314
661	0.062500
666	0.062422
667	0.062500
668	0.062500
669	0.062422
670	0.062422
671	0.062500
672	0.062500
673	0.062577
674	0.062500
675	0.062422
676	0.062500

677	0.031250
678	0.062578
689	0.062500
690	0.062500
691	0.062500
692	0.062500
693	0.062500
694	0.062500
695	0.062500
696	0.062500
697	0.062500
698	0.062500
699	0.062500
700	0.062500
701	0.062500
702	0.062500
703	0.062500
704	0.062500
705	0.062500
706	0.062500
707	0.062500
708	0.062500
709	0.062500
710	0.062500
711	0.062500
712	0.062500
713	0.062500
714	0.062500
715	0.062422
716	0.062500
717	0.062500
718	0.062422
719	0.062422
720	0.062500
721	0.062500
722	0.062578
723	0.062500
724	0.062422
725	0.062500
726	0.031250
727	0.062578
728	0.062423
729	0.062500
730	0.062500
731	0.062423
732	0.062423
733	0.062500
734	0.062500
735	0.062578
736	0.062500
737	0.062423
738	0.062500
739	0.031250
740	0.062578
741	0.062500
742	0.062500

743	0.062500
744	0.062500
745	0.062422
746	0.062422
747	0.060059
750	0.062500
751	0.031211
752	0.031211
753	0.027344
756	0.031250
761	0.062500
762	0.062500
763	0.031250
764	0.031250
765	0.062500
766	0.062500
767	0.062500
768	0.062500
769	0.062500
770	0.062500
771	0.062500
772	0.062500
773	0.062500
774	0.062500
775	0.062500
776	0.062500
777	0.062500
778	0.062500
779	0.062500
780	0.062500
781	0.031250
789	0.033627
790	0.029606
791	0.025553
792	0.021484
793	0.049951
794	0.045898
795	0.041829
796	0.031250
797	0.031250
798	0.031250
799	0.031250
800	0.031250
801	0.031250
802	0.031250
803	0.031250
804	0.031250
805	0.031250
806	0.031250
807	0.031250
808	0.031250
809	0.031250
810	0.031250
811	0.031250
812	0.015625
830	0.062423

831	0.062500
832	0.062500
833	0.062423
834	0.062423
835	0.062500
836	0.062500
837	0.062578
838	0.062500
839	0.062423
840	0.062500
841	0.031250
842	0.062578
843	0.031211
844	0.031250
845	0.031250
846	0.031211
847	0.031211
848	0.031250
849	0.031250
850	0.031289
851	0.031250
852	0.031211
853	0.031250
854	0.015625
855	0.031289
856	0.062500
857	0.062500
858	0.031250
859	0.031250
860	0.062500
861	0.031250
862	0.031250
863	0.015625
TOTAL	15.747211

```
233. ********************************************************************************
234. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
235. SELFWEIGHT Y -1
236. ELEMENT LOAD
237. * LOAD DUE TO BACKFILL
238. 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
239. 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
240. 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
241. 796 TO 801 804 TO 823 840 TO 871 PR GY -30
242. * LOAD DUE TO SURCHARGE
243. 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 431 432 465 TO 474 -
244. 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 TO 651 654 -
245. 659 TO 670 683 TO 685 692 693 695 TO 749 751 TO 753 756 761 TO 780 -
246. 796 TO 801 804 TO 823 840 TO 871 PR GY -20
247. JOINT LOAD
248. 128 FX 9.664 FY -72.134 FZ 1.329
249. ********************************************************************************
250. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
251. JOINT LOAD
252. 128 FX 30.441 FY -263.577 FZ 2.183
253. ********************************************************************************
254. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
```

```
255. JOINT LOAD
256. 128 FX 66.861 FY -128.014 FZ 4.007
257. ********************************************************************************
258. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
259. JOINT LOAD
260. 128 FX -66.861 FY 128.014 FZ -4.007
261. ********************************************************************************
262. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
263. JOINT LOAD
264. 128 FX 6.718 FY -227.583 FZ 96.675
265. ********************************************************************************
266. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
267. JOINT LOAD
268. 128 FX -6.718 FY 227.583 FZ 96.675
269. ********************************************************************************
270. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
271. LOAD COMB 101 1.4DL
272. 1 1.4
273. LOAD COMB 102 1.4DL + 1.7LL
274. 1 1.4 2 1.7
275. LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
276. 1 1.4 3 1.6 2 1.0
277. LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
278. 1 1.4 4 1.6 2 1.0
279. LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
280. 1 1.4 5 1.6 2 1.0
281. LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
282. 1 1.4 6 1.6 2 1.0
283. LOAD COMB 107 0.9DL + 1.6WLX1
284. 1 0.9 3 1.6
285. LOAD COMB 108 0.9DL + 1.6WLX2
286. 1 0.9 4 1.6
287. LOAD COMB 109 0.9DL + 1.6WLZ1
288. 1 0.9 5 1.6
289. LOAD COMB 110 0.9DL + 1.6WLZ2
290. 1 0.9 6 1.6
291. ********************************************************************************
292. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
293. LOAD COMB 201 1.0DL
294. 1 1.0
295. LOAD COMB 202 1.0DL + 1.0LL
296. 1 1.0 2 1.0
297. LOAD COMB 203 1.0DL + 1.0WLX1
298. 1 1.0 3 1.0
299. LOAD COMB 204 1.0DL + 1.0WLX2
300. 1 1.0 4 1.0
301. LOAD COMB 205 1.0DL + 1.0WLZ1
302. 1 1.0 5 1.0
303. LOAD COMB 206 1.0DL + 1.0WLZ2
304. 1 1.0 6 1.0
305. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
306. 1 1.0 3 0.75 2 0.75
307. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
308. 1 1.0 4 0.75 2 0.75
309. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
310. 1 1.0 5 0.75 2 0.75
```



```
311. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
312. 1 1.0 6 0.75 2 0.75
313. LOAD COMB 211 0.6DL + 1.0WLX1
314. 1 0.6 3 1.0
315. LOAD COMB 212 0.6DL + 1.0WLX2
316. 1 0.6 4 1.0
317. LOAD COMB 213 0.6DL + 1.0WLZ1
318. 1 0.6 5 1.0
319. LOAD COMB 214 0.6DL + 1.0WLZ2
320. 1 0.6 6 1.0
321. ********************************************************************************
322. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS        284  NUMBER OF MEMBERS       1
  NUMBER OF PLATES        257  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    283


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=   215/    22/     69 DOF
TOTAL      PRIMARY LOAD CASES =     6, TOTAL DEGREES OF FREEDOM =     855
TOTAL LOAD COMBINATION  CASES =    24  SO FAR.
  SIZE OF STIFFNESS MATRIX =       59 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     14.0/ 146356.5 MB



323. LOAD LIST 101 TO 110
324. START CONCRETE DESIGN
```

```
325. CODE ACI
326. TRACK 2 ALL
327. FC 35000 MEMB 185
328. CLB 0.05 MEMB 185
329. CLS 0.05 ALL
330. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 341 TO 348 351 352 383 TO 429 -
331. 431 432 465 TO 474 607 TO 609 615 TO 619 625 TO 631 635 TO 640 645 646 649 -
332. 650 TO 651 654 659 TO 670 695 TO 749 751 TO 753 756 761 TO 780 808 TO 823 -
333. 840 TO 871
334. MAXMAIN 25 ALL
335. MINMAIN 25 ALL
336. MINSEC 10 ALL
337. DESIGN COLUMN 185
```



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================================================================

     COLUMN  NO.   185  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 700.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  4900.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

 12 - 25 MM               1.202      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
16839.05  13471.24   6253.35   1726.04     276.0
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  741.07  -2436.79    155.37      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
16839.05  13471.24   6253.35   1726.04     276.0
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  741.07  -2436.79    155.37      0.00    0.00000
 ---------------------------------------------------------
                        Pn       Mn       Pn       Mn    (@ Z )
          |            12434.99  1145.49  6217.49  1724.81
       P0 |*           11398.74  1312.93  5181.25  1667.55
          | *          10362.49  1450.83  4145.00  1582.87
    Pn,max|__*          9326.24  1556.76  3108.75  1440.00
          |   *        8289.99  1634.97  2072.50  1252.81
    Pn    |    *       7253.74  1688.98  1036.25  1023.18
   NOMINAL|     *        Pn       Mn       Pn       Mn    (@ Y )
     AXIAL|      *     12434.99  1145.49  6217.49  1724.81
COMPRESSION|       *    11398.74  1312.93  5181.25  1667.55
        Pb|-------*Mb  10362.49  1450.83  4145.00  1582.87
          |      *      9326.24  1556.76  3108.75  1440.00
__________|____*_______ 8289.99  1634.97  2072.50  1252.81
          |  * M0   Mn, 7253.74  1688.98  1036.25  1023.18
          | *   BENDING
    P-tens|*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************
```

338. END CONCRETE DESIGN
339. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 12:33:12 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    MAY 22, 2020                *
            *           Time=    12:33:45                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2 _STABILITY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F2_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 911.85 3.175
   14. MEMBER INCIDENCES
   15. 185 2 128
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 PRIS YD 0.7 ZD 0.7
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FX 9.664 FY -72.134 FZ 1.329
   37. ******************************************************************************
   38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
```



```
   39. JOINT LOAD
   40. 128 FX 30.441 FY -263.577 FZ 2.183
   41. ******************************************************************************
   42. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
   43. JOINT LOAD
   44. 128 FX 66.861 FY -128.014 FZ 4.007
   45. ******************************************************************************
   46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
   47. JOINT LOAD
   48. 128 FX -66.861 FY 128.014 FZ -4.007
   49. ******************************************************************************
   50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
   51. JOINT LOAD
   52. 128 FX 6.718 FY -227.583 FZ 96.675
   53. ******************************************************************************
   54. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
   55. JOINT LOAD
   56. 128 FX -6.718 FY 227.583 FZ 96.675
   57. ******************************************************************************
   58. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   59. LOAD COMB 201 1.0DL
   60. 1 1.0
   61. LOAD COMB 202 1.0DL + 1.0LL
   62. 1 1.0 2 1.0
   63. LOAD COMB 203 1.0DL + 1.0WLX1
   64. 1 1.0 3 1.0
   65. LOAD COMB 204 1.0DL + 1.0WLX2
   66. 1 1.0 4 1.0
   67. LOAD COMB 205 1.0DL + 1.0WLZ1
   68. 1 1.0 5 1.0
   69. LOAD COMB 206 1.0DL + 1.0WLZ2
   70. 1 1.0 6 1.0
   71. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
   72. 1 1.0 3 0.75 2 0.75
   73. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
   74. 1 1.0 4 0.75 2 0.75
   75. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
   76. 1 1.0 5 0.75 2 0.75
   77. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
   78. 1 1.0 6 0.75 2 0.75
   79. LOAD COMB 211 0.6DL + 1.0WLX1
   80. 1 0.6 3 1.0
   81. LOAD COMB 212 0.6DL + 1.0WLX2
   82. 1 0.6 4 1.0
   83. LOAD COMB 213 0.6DL + 1.0WLZ1
   84. 1 0.6 5 1.0
   85. LOAD COMB 214 0.6DL + 1.0WLZ2
   86. 1 0.6 6 1.0
   87. ******************************************************************************
   88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 146353.1 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 22,2020 TIME= 12:33:49 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 7 STAAD OUTPUT – FOUNDATION F3

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    MAY 22, 2020                 *
*          Time=    12:37: 8                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3 DESIGN.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F3
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 6 0.75 909.9 3.175; 34 -0.5 909.9 3.175; 35 -0.25 909.9 3.175
   14. 41 0.25 909.9 3.175; 42 0.5 909.9 3.175; 128 0 911.85 3.175
   15. 302 -0.5 909.9 3.42438; 304 -0.25 909.9 3.42438; 305 0 909.9 3.42438
   16. 306 0.25 909.9 3.42438; 307 0.5 909.9 3.42438; 308 0.75 909.9 3.42438
   17. 309 1 909.9 3.175; 311 1.275 909.9 3.175; 313 1 909.9 3.42438
   18. 314 1.275 909.9 3.42438; 322 -0.5 909.9 3.67438; 323 -0.25 909.9 3.67438
   19. 324 0 909.9 3.67438; 325 0.25 909.9 3.67438; 326 0.5 909.9 3.67438
   20. 327 0.75 909.9 3.67438; 328 1 909.9 3.67438; 329 1.275 909.9 3.67438
   21. 399 -0.5 909.9 2.67438; 400 -0.25 909.9 2.67438; 401 0 909.9 2.67438
   22. 402 0.25 909.9 2.67438; 403 0.5 909.9 2.67438; 404 0.75 909.9 2.67438
   23. 405 1 909.9 2.67438; 406 1.275 909.9 2.67438; 410 -0.5 909.9 2.92438
   24. 411 -0.25 909.9 2.92438; 412 0 909.9 2.92438; 413 0.25 909.9 2.92438
   25. 414 0.5 909.9 2.92438; 415 0.75 909.9 2.92438; 416 1 909.9 2.92438
   26. 417 1.275 909.9 2.92438; 420 1.525 909.9 3.42438; 421 1.525 909.9 3.175
   27. 422 1.525 909.9 3.67438; 429 1.525 909.9 2.67438; 430 1.525 909.9 2.92438
   28. 468 1.775 909.9 3.42438; 469 1.775 909.9 3.175; 470 1.775 909.9 3.67438
   29. 477 1.775 909.9 2.67438; 478 1.775 909.9 2.92438; 479 1.275 911.85 3.175
   30. MEMBER INCIDENCES
   31. 185 2 128; 487 311 479
   32. ELEMENT INCIDENCES SHELL
   33. 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2; 333 306 307 42 41
   34. 334 307 308 6 42; 337 308 313 309 6; 338 313 314 311 309; 344 322 323 304 302
   35. 345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306
   36. 348 326 327 308 307; 349 327 328 313 308; 350 328 329 314 313
   37. 414 410 411 400 399; 415 411 412 401 400; 416 412 413 402 401
   38. 417 413 414 403 402; 418 414 415 404 403; 419 415 416 405 404
```



```
   39. 420 416 417 406 405; 424 34 35 411 410; 425 35 2 412 411; 426 2 41 413 412
   40. 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415; 430 309 311 417 416
   41. 433 314 420 421 311; 434 329 422 420 314; 441 417 430 429 406
   42. 442 311 421 430 417; 477 420 468 469 421; 478 422 470 468 420
   43. 485 430 478 477 429; 486 421 469 478 430
   44. START GROUP DEFINITION
   45. ELEMENT
   46. _RAFT 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 -
   47. 441 442 477 478 485 486
   48. _RAFT_MOM 327 333 334 337 344 TO 350 414 TO 420 424 427 TO 429 434 441 477 -
   49. 478 485 486
   50. _RAFT_SHEAR 327 333 334 337 344 TO 350 414 TO 420 424 427 TO 429 434 441 477 -
   51. 478 485 486
   52. MEMBER
   53. END GROUP DEFINITION
   54. ELEMENT PROPERTY
   55. 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
   56. 477 478 485 486 THICKNESS 0.5
   57. DEFINE MATERIAL START
   58. ISOTROPIC CONCRETE
   59. E 2.17185E+007
   60. POISSON 0.17
   61. DENSITY 25
   62. ALPHA 1E-005
   63. DAMP 0.05
   64. TYPE CONCRETE
   65. STRENGTH FCU 27579
   66. END DEFINE MATERIAL
   67. MEMBER PROPERTY AMERICAN
   68. 185 487 PRIS YD 0.5 ZD 0.5
   69. CONSTANTS
   70. MATERIAL CONCRETE ALL
   71. SUPPORTS
   72. 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
   73. 477 478 485 486 PLATE MAT DIRECT Y SUBGRADE 45600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.062500
34	0.031250
35	0.062500
41	0.062500
42	0.062500
302	0.031211
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.062422
309	0.065625
311	0.065625
313	0.065544
314	0.065544
322	0.015625

323	0.031250
324	0.031250
325	0.031250
326	0.031250
327	0.031250
328	0.032812
329	0.032812
399	0.015625
400	0.031250
401	0.031250
402	0.031250
403	0.031250
404	0.031250
405	0.032813
406	0.032813
410	0.031289
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.062578
416	0.065706
417	0.065706
420	0.062422
421	0.062500
422	0.031250
429	0.031250
430	0.062578
468	0.031211
469	0.031250
470	0.015625
477	0.015625
478	0.031289
TOTAL	2.275000

```
74. ********************************************************************************
75. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
76. SELFWEIGHT Y -1
77. ELEMENT LOAD
78. * LOAD DUE TO BACKFILL
79. 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
80. 477 478 485 486 PR GY -15
81. * LOAD DUE TO SURCHARGE
82. 327 328 332 TO 334 337 338 344 TO 350 414 TO 420 424 TO 430 433 434 441 442 -
83. 477 478 485 486 PR GY -20
84. JOINT LOAD
85. 479 FX 0.131 FY -3.735 FZ -0.041
86. 128 FX -0.131 FY -3.507 FZ -0.121
87. ********************************************************************************
88. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
89. JOINT LOAD
90. 479 FX 0.43 FY -11.123 FZ -0.168
91. 128 FX -0.43 FY -10.819 FZ -0.501
92. ********************************************************************************
93. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
94. JOINT LOAD
95. 479 FX 1.349 FY -2.444 FZ 0
```



```
96. 128 FX 3.963 FY 2.444 FZ 0.003
97. ********************************************************************************
98. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
99. JOINT LOAD
100. 479 FX -1.349 FY 2.444 FZ 0
101. 128 FX -3.963 FY -2.444 FZ -0.003
102. ********************************************************************************
103. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
104. JOINT LOAD
105. 479 FX 0.063 FY 0.218 FZ 0.603
106. 128 FX 0.199 FY 0.455 FZ 0.786
107. ********************************************************************************
108. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
109. JOINT LOAD
110. 479 FX -0.063 FY -0.218 FZ -0.603
111. 128 FX -0.199 FY -0.455 FZ -0.786
112. ********************************************************************************
113. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
114. LOAD COMB 101 1.4DL
115. 1 1.4
116. LOAD COMB 102 1.4DL + 1.7LL
117. 1 1.4 2 1.7
118. LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
119. 1 1.4 3 1.6 2 1.0
120. LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
121. 1 1.4 4 1.6 2 1.0
122. LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
123. 1 1.4 5 1.6 2 1.0
124. LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
125. 1 1.4 6 1.6 2 1.0
126. LOAD COMB 107 0.9DL + 1.6WLX1
127. 1 0.9 3 1.6
128. LOAD COMB 108 0.9DL + 1.6WLX2
129. 1 0.9 4 1.6
130. LOAD COMB 109 0.9DL + 1.6WLZ1
131. 1 0.9 5 1.6
132. LOAD COMB 110 0.9DL + 1.6WLZ2
133. 1 0.9 6 1.6
134. ********************************************************************************
135. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
136. LOAD COMB 201 1.0DL
137. 1 1.0
138. LOAD COMB 202 1.0DL + 1.0LL
139. 1 1.0 2 1.0
140. LOAD COMB 203 1.0DL + 1.0WLX1
141. 1 1.0 3 1.0
142. LOAD COMB 204 1.0DL + 1.0WLX2
143. 1 1.0 4 1.0
144. LOAD COMB 205 1.0DL + 1.0WLZ1
145. 1 1.0 5 1.0
146. LOAD COMB 206 1.0DL + 1.0WLZ2
147. 1 1.0 6 1.0
148. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
149. 1 1.0 3 0.75 2 0.75
150. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
151. 1 1.0 4 0.75 2 0.75
```

```
152. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
153. 1 1.0 5 0.75 2 0.75
154. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
155. 1 1.0 6 0.75 2 0.75
156. LOAD COMB 211 0.6DL + 1.0WLX1
157. 1 0.6 3 1.0
158. LOAD COMB 212 0.6DL + 1.0WLX2
159. 1 0.6 4 1.0
160. LOAD COMB 213 0.6DL + 1.0WLZ1
161. 1 0.6 5 1.0
162. LOAD COMB 214 0.6DL + 1.0WLZ2
163. 1 0.6 6 1.0
164. ********************************************************************************
165. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 52 NUMBER OF MEMBERS 2
NUMBER OF PLATES 36 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 50

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 38/ 10/ 36 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 162
TOTAL LOAD COMBINATION CASES = 24 SO FAR.
SIZE OF STIFFNESS MATRIX = 6 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 146353.4 MB

```
166. LOAD LIST 101 TO 110
167. START CONCRETE DESIGN
```



```
168. CODE ACI
169. TRACK 2 ALL
170. FC 35000 MEMB 185 487
171. CLB 0.05 MEMB 185 487
172. CLS 0.05 ALL
173. CLT 0.05 MEMB 185 487
174. MAXMAIN 25 ALL
175. MINMAIN 25 ALL
176. MINSEC 10 ALL
177. DESIGN COLUMN 185 487
```

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

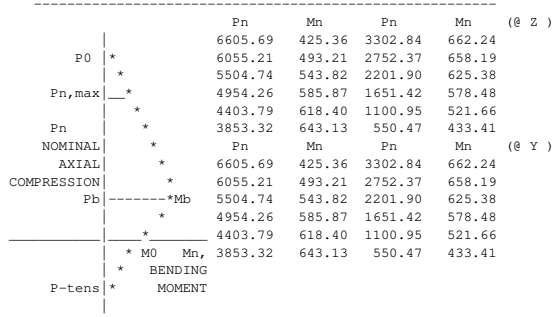
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	7.55	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	7.55	0.00	0.00000



COLUMN NO. 487 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 500.0 X 500.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	8.04	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8945.20	7156.16	3013.84	670.78	222.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
333.62	-1624.53	8.04	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

```
178. END CONCRETE DESIGN
179. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= MAY 22,2020   TIME= 12:37:12 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
```

```
            **************************************************
            *                                                *
            *           STAAD.Pro V8i SELECTseries5          *
            *           Version  20.07.10.64                 *
            *           Proprietary Program of               *
            *           Bentley Systems, Inc.                *
            *           Date=    MAY 22, 2020                *
            *           Time=    12:38:52                    *
            *                                                *
            *      USER ID: LNT ECC DIV                      *
            **************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3 _STABILTY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F3-STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 909.9 3.175; 128 0 911.85 3.175; 311 1.275 909.9 3.175
   14. 479 1.275 911.85 3.175
   15. MEMBER INCIDENCES
   16. 185 2 128; 487 311 479
   17. DEFINE MATERIAL START
   18. ISOTROPIC CONCRETE
   19. E 2.17185E+007
   20. POISSON 0.17
   21. DENSITY 25
   22. ALPHA 1E-005
   23. DAMP 0.05
   24. TYPE CONCRETE
   25. STRENGTH FCU 27579
   26. END DEFINE MATERIAL
   27. MEMBER PROPERTY AMERICAN
   28. 185 487 PRIS YD 0.5 ZD 0.5
   29. CONSTANTS
   30. MATERIAL CONCRETE ALL
   31. SUPPORTS
   32. 2 311 FIXED
   33. ******************************************************************************
   34. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
             MASTER/SLAVE OR IF UNCONNECTED JOINTS.
   35. SELFWEIGHT Y -1
   36. JOINT LOAD
```



```
   37. 479 FX 0.131 FY -3.735 FZ -0.041
   38. 128 FX -0.131 FY -3.507 FZ -0.121
   39. ******************************************************************************
   40. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
   41. JOINT LOAD
   42. 479 FX 0.43 FY -11.123 FZ -0.168
   43. 128 FX -0.43 FY -10.819 FZ -0.501
   44. ******************************************************************************
   45. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
   46. JOINT LOAD
   47. 479 FX 1.349 FY -2.444 FZ 0
   48. 128 FX 3.963 FY 2.444 FZ 0.003
   49. ******************************************************************************
   50. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
   51. JOINT LOAD
   52. 479 FX -1.349 FY 2.444 FZ 0
   53. 128 FX -3.963 FY -2.444 FZ -0.003
   54. ******************************************************************************
   55. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
   56. JOINT LOAD
   57. 479 FX 0.063 FY 0.218 FZ 0.603
   58. 128 FX 0.199 FY 0.455 FZ 0.786
   59. ******************************************************************************
   60. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
   61. JOINT LOAD
   62. 479 FX -0.063 FY -0.218 FZ -0.603
   63. 128 FX -0.199 FY -0.455 FZ -0.786
   64. ******************************************************************************
   65. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   66. LOAD COMB 201 1.0DL
   67. 1 1.0
   68. LOAD COMB 202 1.0DL + 1.0LL
   69. 1 1.0 2 1.0
   70. LOAD COMB 203 1.0DL + 1.0WLX1
   71. 1 1.0 3 1.0
   72. LOAD COMB 204 1.0DL + 1.0WLX2
   73. 1 1.0 4 1.0
   74. LOAD COMB 205 1.0DL + 1.0WLZ1
   75. 1 1.0 5 1.0
   76. LOAD COMB 206 1.0DL + 1.0WLZ2
   77. 1 1.0 6 1.0
   78. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
   79. 1 1.0 3 0.75 2 0.75
   80. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
   81. 1 1.0 4 0.75 2 0.75
   82. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
   83. 1 1.0 5 0.75 2 0.75
   84. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
   85. 1 1.0 6 0.75 2 0.75
   86. LOAD COMB 211 0.6DL + 1.0WLX1
   87. 1 0.6 3 1.0
   88. LOAD COMB 212 0.6DL + 1.0WLX2
   89. 1 0.6 4 1.0
   90. LOAD COMB 213 0.6DL + 1.0WLZ1
   91. 1 0.6 5 1.0
   92. LOAD COMB 214 0.6DL + 1.0WLZ2
```

```
93. 1 0.6 6 1.0
94. ********************************************************************************
95. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          4  NUMBER OF MEMBERS       2
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      2


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
TOTAL      PRIMARY LOAD CASES =      6, TOTAL DEGREES OF FREEDOM =      12
TOTAL LOAD COMBINATION  CASES =     14  SO FAR.
  SIZE OF STIFFNESS MATRIX =        1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/ 146352.7 MB



96. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= MAY 22,2020   TIME= 12:38:57 ****
```



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     ************************************************************
     *   For technical assistance on STAAD.Pro, please visit    *
     *   http://selectservices.bentley.com/en-US/               *
     *                                                          *
     *   Details about additional assistance from               *
     *   Bentley and Partners can be found at program menu      *
     *   Help->Technical Support                                *
     *                                                          *
     *      Copyright (c) 1997-2014 Bentley Systems, Inc.
     *               http://www.bentley.com                     *
     ************************************************************
```